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Review

Vision transformers in precision agriculture: A comprehensive survey

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ABSTRACT

Detecting plant diseases is a crucial aspect of modern agriculture, playing a key role in maintaining crop health and ensuring sustainable yields. Traditional approaches, though still valuable, often rely on manual inspection or conventional machine learning (ML) techniques, both of which face limitations in scalability and accuracy. The emergence of Vision Transformers (ViTs) marks a significant shift in this landscape by enabling superior modeling of long-range dependencies and offering improved scalability for complex visual tasks. This survey provides a rigorous and structured analysis of impactful studies that employ ViT-based models, along with a comprehensive categorization of existing research. It also offers a quantitative synthesis of reported performance — with accuracies ranging from 75.00% to 100.00% — highlighting clear trends in model effectiveness and identifying consistently high-performing architectures. In addition, this study examines the inductive biases of CNNs and ViTs, which is the first analysis of these architectural priors within an agricultural context. Further contributions include a comparative taxonomy of prior studies, an evaluation of dataset limitations and metric inconsistencies, and a statistical assessment of model efficiency across diverse crop-image sources. Collectively, these efforts clarify the current state of the field, identify critical research gaps, and outline key challenges — such as data diversity, interpretability, computational cost, and field adaptability — that must be addressed to advance the practical deployment of ViT technologies in precision agriculture.

1. Introduction

Plant diseases pose a significant challenge to global agriculture, affecting crop yields, food security, and the economic well-being of farmers. Early and accurate detection is essential to effectively manage these diseases and minimize their impact (Buja et al., 2021). Traditionally, the diagnosis of plant diseases has depended on labor-intensive manual inspections by experts or on conventional Machine Learning (ML) techniques, both of which have inherent limitations. Manual inspection is not only time-consuming but also subject to human error and inconsistency, making it impractical for large-scale agricultural operations (Garg et al., 2024; Mahlein, 2016). Conventional ML methods are largely dependent on manual feature extraction, which limits their capacity to generalize across diverse environmental conditions, such as variations in lighting, background, and plant species. Moreover, these techniques are unable to effectively capture complex spatial patterns directly from raw image data (Ferentinos, 2018). To address these limitations, researchers are increasingly turning to automated solutions that harness recent advances in ML and Computer Vision (CV).

Convolutional Neural Network (CNN), a widely used traditional Deep Learning (DL) method, has shown some success in automating the disease detection process (Sladojevic et al., 2016). This has created a growing demand for more robust, scalable, and intelligent systems for plant disease detection. In response, several studies have explored the potential of DL to meet these challenges. Some existing surveys provide general overviews of DL approaches, highlighting their advantages over traditional methods and outlining the progress made in applying these techniques to plant disease detection tasks. For example, one study (Shoaib et al., 2023) reviews how DL and ML techniques have been employed for plant disease detection from image data. The study describes the transition from handcrafted features with the traditional DL methods to deep models like CNNs and segmentation models. Another work (Jackulin & Murugavalli, 2022) discusses the application of ML and DL techniques in the early detection and classification of plant diseases, highlighting their importance in improving agricultural yield and reducing crop loss. The contrast between classical ML and DL methods shows that DL outperforms ML in terms of accuracy and scalability. This study emphasizes the importance of addressing current

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deficiencies and proposes DL as a promising direction for creating automatic, systematic, and scalable plant disease diagnostic systems that support both scientists and farmers in agricultural decision-making. Another study (Ahmad et al., 2023) explores the use of DL techniques for plant disease detection and management in agriculture. It addresses key challenges including dataset quality, imaging sensors, model generalizability, estimation of severity, and performance comparison with human capabilities. Commonly used datasets are categorized, along with their limitations in practical scenarios. The study outlines various DL techniques, with a particular focus on CNN-based models, employed for image classification, object detection, and semantic segmentation. It highlights limitations in the ability of existing models to generalize to field conditions, primarily due to overfitting and imbalanced data. A separate study (Li et al., 2021) explores the application of DL in plant disease recognition and classification, emphasizing its critical role in agricultural productivity and food security. It illustrates how DL improves upon traditional image processing techniques through automated feature extraction and enhanced objectivity and efficiency. The paper discusses foundational concepts, model performance criteria, and the evolution of DL in plant disease recognition. It also addresses the need for data augmentation to manage limited datasets and the use of visualization methods to improve model interpretability. The study concludes with future challenges, such as the need for more varied datasets, enhanced model robustness, and the integration of hyperspectral imaging for early detection.

Recent advancements in DL have demonstrated the potential of CNNs and their variants in accurately detecting and classifying plant diseases. However, despite their success, CNN-based models often struggle to capture global contextual information due to their inherent inductive biases and locality constraints. These limitations underscore the need for models that can overcome such constraints, thereby motivating researchers to explore and adapt architectures originally developed in other domains to address these challenges.

The introduction of the Transformer architecture has transformed the field of Natural Language Processing (NLP), enabling significant advancements in machine translation, text summarization, and language modeling, and setting new performance benchmarks across a wide range of language understanding tasks (Radford et al., 2019). By leveraging parallel processing and self-attention mechanisms, Transformers excel at capturing long-range dependencies and contextual relationships. This breakthrough has inspired researchers to explore the application of Transformers in other domains, including CV. It was soon recognized that the same properties that made Transformers effective in NLP, namely their ability to model long-range dependencies and contextually relevant information, could also be advantageous for image analysis tasks (Dosovitskiy, 2020). Unlike CNNs, which rely on inductive biases such as local connectivity and spatial hierarchies, Transformers offer a more flexible architecture with fewer assumptions about input structure. This flexibility enables them to learn rich, complex patterns directly from visual data, making them a promising alternative for various CV applications, including automated plant disease detection.

The use of Vision Transformers (ViTs) in agriculture is a novel and rapidly emerging research area. To the best of our knowledge, there are a few works whose focus is on ViTs; however, most of them mention it only briefly and do not provide a comprehensive analysis of their applications in precision agriculture. For example, a review (Pacal et al., 2024) categorizes DL techniques applied to plant disease detection, organizing the literature into three tasks: classification, detection, and segmentation of plant diseases from leaf images. The review evaluates a broad range of DL models, with a strong focus on CNNs such as ResNet (He et al., 2016a), VGG (Simonyan & Zisserman, 2014), and EfficientNet (Tan & Le, 2019), alongside growing interest in ViTs. Another study (Upadhyay et al., 2025) examines the use of DL and CV in plant disease detection, systematically analyzing the transition from traditional image processing to modern DL-based approaches. It assesses the suitability and performance of image processing techniques like thresholding, edge detection, and region-based segmentation in real-world agriculture. The study categorizes various DL architectures, such as CNNs, GAN, ViTs, and Vision–Language Models, and highlights their scalability and potential for practical implementation. However, while ViTs are introduced, the review lacks a deep evaluation of specific architectures or performance benchmarks in agricultural contexts.

In this work, we examine the use of ViT in agriculture, aiming to identify how this architecture has been adapted for applications such as crop classification, disease detection, yield prediction, and precision agriculture. Given ViT's success in CV across multiple domains, its potential to transform agricultural image analysis is significant. By reviewing existing literature, we aim to uncover current trends, assess model performance, and highlight challenges unique to agricultural datasets and environments. This exploration lays a foundation for future studies and contributes to building a structured body of knowledge at the intersection of DL and agriculture.

To achieve this, we conducted a comprehensive literature search using DBLP, Google Scholar, arXiv Sanity Preserver, IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and other relevant academic databases. Tailored search queries were designed to systematically retrieve scholarly publications on hybrid CNN–ViT models for precision agriculture, plant disease detection, and related agricultural stress analysis.

Our search queries were designed around multiple keyword clusters, including:

- **Model/Method Keywords:** "CNN", "convolutional neural network", "MobileNet", "ViT", "vision transformer", "transformer", "Swin Transformer", "DeiT", "MobileViT", "SwinT", "hybrid model", "CNN–ViT", "deep learning", "attention mechanism", "feature fusion", and "multi-modal deep learning".
- **Agriculture Context Keywords:** "precision agriculture", *agricultur**, *crop**, "smart farming", "plant health", "plant phenotyping", "field phenotyping", "yield prediction", "agroinformatics", "weed detection", "fruit ripeness", and "crop monitoring".
- **Plant Disease/Stress Keywords:** "plant disease", "crop disease", "leaf disease", "disease detection", "disease classification", "pest detection", "plant stress".
- **Imaging/Modalities Keywords:** image*, vision, "hyperspectral imaging", multispectral, "thermal imaging", RGB, UAV, drone*, "remote sensing", "field imagery", and "smartphone-based detection".

An example Boolean search string combining these terms is as follows:

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(CNN OR ''convolutional neural network'' OR MobileNet) AND (ViT OR ''vision transformer'' OR transformer OR ''Swin Transformer'' OR DeiT OR SwinT OR MobileViT OR ''hybrid model'' OR ''CNN--ViT'') AND (''precision agriculture'' OR agricultur* OR crop* OR ''plant health'' OR ''plant phenotyping'' OR ''yield prediction'' OR ''smart farming'' OR agroinformatics) AND (''plant disease'' OR ''disease detection'' OR ''disease classification'' OR pest* OR ''nutrient deficiency'' OR ''plant stress'') AND (image* OR vision OR ''hyperspectral imaging'' OR multispectral OR ''thermal imaging'' OR RGB OR UAV OR drone* OR ''remote sensing'' OR ''field imagery'' OR ''smartphone-based detection'')
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To ensure the relevance of the retrieved literature, each study was carefully assessed with respect to its novelty, methodological contributions, and overall impact, with particular emphasis on pioneering research employing hybrid CNN–ViT architectures for plant disease detection and classification. Studies that offered rigorous benchmarking, multi-crop validation, or practical field-level implementation were also given

Table 1
Comparison of existing surveys related to deep learning and ViT in agriculture.

Study	Main Focus	Quantitative Analysis	Primary Focus on ViTs	Discussion of Inductive Bias
Li et al. (2021)	Review of research progress of DL technology in crop leaf disease identification	✓	✗	✗
Jackulin and Murugavalli (2022)	Review of ML and DL approaches for plant disease detection	✓	✗	✗
Shoaib et al. (2023)	Review of ML/DL techniques for plant disease detection	✓	✗	✗
Ahmad et al. (2023)	Overview of DL applications for plant disease diagnosis and management	✓	✗	✗
Pacal et al. (2024)	Systematic review of DL techniques for plant disease detection	✓	✗	✗
Upadhyay et al. (2025)	Review of DL and CV techniques and models for plant disease detection	✓	✗	✗
This work	Comprehensive survey on the application of ViTs in precision agriculture	✓	✓	✓

priority. Applying these criteria, the most significant publications were selected for detailed analysis. It should be noted that, although some relevant studies may not have been included, the primary aim was to provide a comprehensive synthesis of the most influential and impactful research on the application of Transformer-based models in precision agriculture. Table 1 highlights the unique contributions of this study compared with related works.

In summary, the key contributions of this work are as follows:

- We provide an overview of the Transformer and ViT architectures and their functionality
- We have comprehensively addressed the concept of inductive biases and their role in ML and DL models, especially CNNs and ViTs.
- We present a comprehensive survey on the application of ViTs in the context of precision agriculture, systematically analyzing 50 high-impact peer-reviewed studies.
- We critically analyze the current challenges and unresolved issues in the field, highlight emerging research trends, formulate key open challenges, and suggest promising directions for future investigation.

The structure of this paper is as follows: Section 2 provides an overview of the Transformer architecture, followed by a discussion of the ViT. Section 3 explores the concept of inductive biases in ML and DL, with particular focus on how these biases manifest in CNNs and ViTs. In Section 4, we review the applications of ViTs in precision agriculture, categorizing the models into two main groups: pure and hybrid. Section 5 highlights the key findings and outlines the open challenges identified in the reviewed literature. Finally, Section 6 presents the conclusions of the paper.

2. Background

2.1. Transformer

The Transformer model (Vaswani, 2017) brought a groundbreaking shift to NLP by introducing the self-attention mechanism, which allows the model to evaluate the importance of different words in a sentence during sequence encoding and generation. This innovation overcomes the limitations of earlier architectures, such as Recurrent Neural Networks (RNNs) and CNNs, which often struggled with long-range dependencies and parallelization. By processing all tokens in a sequence simultaneously, self-attention captures long-range relationships more effectively, making the Transformer architecture both efficient and highly scalable for a wide range of sequence modeling tasks. In the following subsection, we examine its main components.

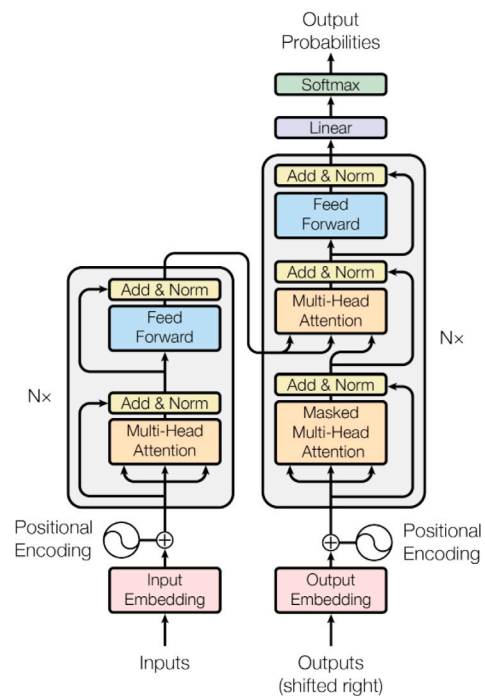

Fig. 1. The Encoder–Decoder architecture of the Transformer model (Vaswani, 2017). The encoder processes the input sequence to generate contextualized representations, which are then passed to the decoder to produce the output sequence. This structure enables efficient parallel processing and captures long-range dependencies through self-attention mechanisms, allowing for better modeling of relationships between distant elements in the sequence.

2.1.1. Encoder–decoder structure

The Transformer architecture comprises an encoder and a decoder, each consisting of multiple layers of self-attention and Feed-Forward Neural Networks. The encoder processes the input sequence into a contextual representation, while the decoder generates the output sequence based on both the encoder's output and previously generated tokens. This encoder–decoder structure is commonly used in tasks such as machine translation, whereas certain variants, such as GPT (Brown et al., 2020), employ only the decoder for autoregressive language modeling. Fig. 1 illustrates the overall architecture of the Transformer's encoder–decoder structure.

Each encoder layer consists of two main sublayers:

1. Self-Attention Mechanism
2. Feed-Forward Neural Network

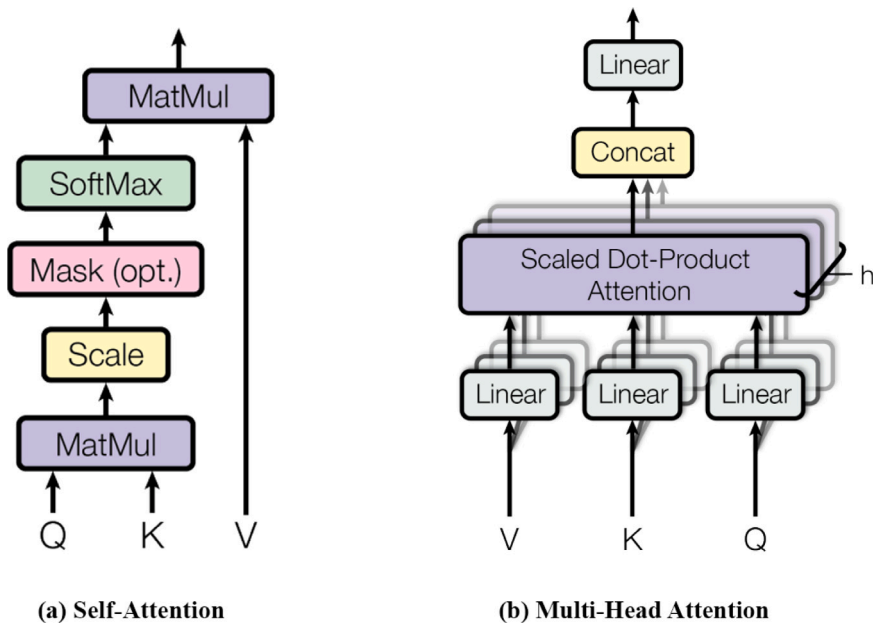

Fig. 2. (a) Self-Attention mechanism. (b) Multi-Head Attention. The attention computation uses queries (Q), keys (K), and values (V) to generate weighted representations, allowing the model to focus on relevant parts of the input sequence based on learned attention scores. Multi-Head Attention runs multiple self-attention operations in parallel, each with different learned projections of Q, K, and V, allowing the model to capture diverse contextual relationships from different representation subspaces.

Each sublayer is followed by a residual connection and layer normalization.

Each decoder layer consists of three sublayers:

1. Masked Self-Attention Mechanism
2. Encoder–Decoder Attention Mechanism
3. Feed-Forward Neural Network

Similar to the encoder, each sublayer is followed by a residual connection and layer normalization.

2.1.2. Self-attention mechanism

The self-attention mechanism is the core innovation of the Transformer architecture, allowing the model to dynamically focus on different parts of the input sequence. As shown in Fig. 2(a), this attention is computed using the following steps:

1. Input Matrices: The inputs consist of queries (Q), keys (K), and values (V), all derived from the input embeddings.
2. Attention Scores: Calculate the dot products of the query (Q) with all keys (K). This results in a matrix of raw attention scores, which determines how much attention each query should pay to the different keys.
3. Scaling: Scale the attention scores by the square root of the dimension of the keys, $\sqrt{d_k}$, to avoid large values that could result in very small gradients during training.
4. Softmax: Apply the softmax function to the scaled attention scores to normalize them, turning them into probabilities (attention weights) that sum to 1.
5. Weighted Sum: Compute the weighted sum of the values (V) using the attention weights. The result is a new set of embeddings (or representations), where each one is a mixture of the original values weighted by how much attention was paid to the corresponding key.

2.1.3. Multi-head attention

To allow the model to focus on different parts of the sequence simultaneously, the Transformer uses multi-head attention, which involves multiple parallel attention layers (heads). The process can be summarized as follows:

1. Linear Projections: Linearly project the queries, keys, and values h times using different learned projection matrices for each head, resulting in h distinct sets of queries, keys, and values.
2. Parallel Attention Heads: Apply the scaled dot-product attention mechanism independently to each set of queries, keys, and values in parallel. This allows the model to attend to different parts of the input sequence from different subspaces.
3. Concatenation: Concatenate the outputs of all h attention heads, forming a single vector of combined information.
4. Final Linear Projection: Apply a final linear transformation to the concatenated output, resulting in the final multi-head attention output that will be passed to subsequent layers.

Fig. 2(b) provides a schematic representation of the multi-head attention mechanism.

2.2. Vision transformer

Inspired by the success of Transformers in NLP, A. Dosovitskiy (2020) adapted the Transformer architecture for CV tasks, leading to the creation of ViT. Before this breakthrough, visual attention mechanisms were typically combined with convolutional networks or used to enhance specific components, while retaining the overall CNN structure. Early attention-based architectures had shown promise in CV (Hu et al., 2018), but their implementation on hardware accelerators often required complex and specialized engineering, as these systems were primarily optimized for convolutional operations. The key insight of ViT was to show that convolutional networks are not essential: a simple Transformer applied directly to sequences of image patches can achieve strong results in image classification. In ViT, images are divided into fixed-size patches, each treated as a token, similar to words in

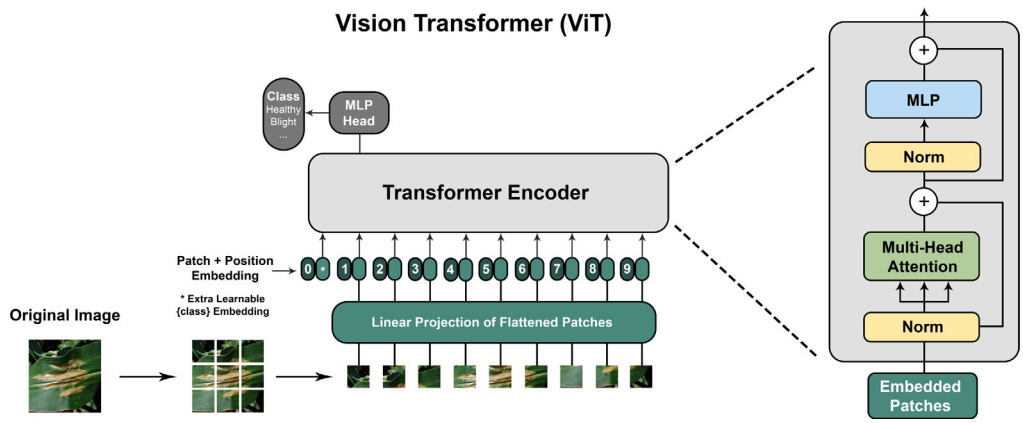

Fig. 3. Vision Transformer architecture (Dosovitskiy, 2020). The image is split into fixed-size patches, which are linearly embedded, combined with positional embeddings, and fed into a Transformer encoder. The model captures global image context through self-attention across all patches.

NLP. These patches are linearly embedded, combined with positional embeddings, and processed through multiple self-attention layers. This design enables ViTs to effectively capture long-range dependencies and contextual relationships within images, leading to performance that often surpasses traditional CNNs on various classification benchmarks. By leveraging the self-attention mechanism of Transformers, ViTs have revolutionized the field of CV. The following section delves into the architecture of ViTs, explaining their key components and operations in detail. Fig. 3 illustrates the ViT architecture, comprising Patch Embedding, Positional Embedding, the Transformer Encoder, and the Classification Head.

2.2.1. Patch embedding

In ViTs, the input image is divided into fixed-size patches. Each patch is then flattened and projected into a lower-dimensional space.

Given an image $X \in \mathbb{R}^{H\times W\times C}$ with height H, width W, and C channels, it is divided into patches of size $P \times P$. Thus, the number of patches N is $\frac{HW}{P^2}$.

Each patch is then linearly transformed into an embedding vector using a trainable projection matrix $E \in \mathbb{R}^{(P^2\cdot C)\times D}$ where D is the dimensionality of the embedding space.

- The embedding dimension D in a ViT plays a crucial role in transforming the flattened image patches into a suitable format for processing by the Transformer architecture.
- An embedding is a vector representation of the input data. For image patches, it means converting the flattened pixel values into a high-dimensional vector that captures more meaningful information.
- Directly using the raw pixel values of image patches may not be efficient or effective for the Transformer model. Embedding helps transform these raw values into a more abstract, dense representation.
- The choice of the embedding dimension D in a ViT is a crucial design decision that balances the model's representational power and computational efficiency.

Let $z_0^P \in \mathbb{R}^D$ denote the initial embedding of the Pth image patch. It is obtained by flattening the patch and projecting it into a D-dimensional space using a learnable linear transformation:

$$z_0^P = \text{Flatten}(x^P)E, \quad \text{for } P = 1 \text{ to } N \tag{1}$$

where $z_0^P \in \mathbb{R}^D$ is the embedding of the Pth patch at Transformer layer-0 (before any attention layers), x^P is the Pth image patch before projection and $E \in \mathbb{R}^{(P^2.C)\times D}$ is a learnable weight matrix that projects each flattened image patch into a D-dimensional embedding space. In the agricultural context, each x^P represents a flattened image patch extracted from a crop or leaf image (e.g., a section of a diseased maize leaf). The embedding z_0^P thus encodes localized texture and color information corresponding to potential disease symptoms or healthy tissue.

2.2.2. Positional embedding

To retain the spatial information, the positional embedding $P \in \mathbb{R}^{N\times D}$ is added to the patch embeddings:

$$z_0 = [x_1E\ ;\ x_2E\ ;\ \dots\ ;\ x_NE] + P \tag{2}$$

where z_0 is the full embedded input sequence after adding positional information at Transformer layer-0 (before any attention layers), $x_i \in \mathbb{R}^{P^2.C}$ is the ith flattened image patch, $E \in \mathbb{R}^{(P^2.C)\times D}$ is the patch embedding projection matrix and $P \in \mathbb{R}^{N\times D}$ is the positional embeddings. The positional embeddings added to the patch embeddings can be either learned or fixed. These positional embeddings help the model preserve the spatial arrangement of patches across a leaf image, ensuring that spatial cues such as lesion position or venation patterns remain distinguishable during learning.

2.2.3. Transformer encoder

The Transformer encoder consists of multiple layers, each composed of a Multi-Head Self-Attention mechanism and a Feed-Forward Neural Network.

1-Multi-Head Self-Attention (MHSA):

The self-attention mechanism computes a weighted sum of the input sequences, focusing on different parts of the sequence for each output element. Given an input sequence $z \in \mathbb{R}^{N\times D}$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{D}}\right)V \tag{3}$$

where $Q = zW_Q$, $K = zW_K$, and $V = zW_V$ are query, key, and value matrices, respectively. W_Q, W_K and W_V are learned projection matrices.

For multi-head attention, multiple attention heads are used:

$$\text{MHSA}(z) = [\text{head}_1\ ;\ \text{head}_2\ ;\ \dots\ ;\ \text{head}_h]W_O \tag{4}$$

where $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$ and $W_O \in \mathbb{R}^{hD\times D}$ is a learned projection matrix. In plant imaging, the attention mechanism enables the model to relate visually distant regions, such as scattered lesions across

a leaf, by assigning higher weights to correlated patches. This helps the ViT capture disease patterns that extend beyond local neighborhoods.

2- Feed-Forward Neural Network (FFN)

Each encoder layer also includes a Feed-Forward Neural Network, applied to each position separately and identically:

$$\mathrm{FFN}(x) = \max(0,\, xW_1 + b_1)W_2 + b_2 \tag{5}$$

where $W_1 \in \mathbb{R}^{D \times D_{ff}}$ and $W_2 \in \mathbb{R}^{D_{ff} \times D}$ are learned weight matrices, $b_1 \in \mathbb{R}^{D_{ff}}$ and $b_2 \in \mathbb{R}^{D}$ are learned bias vectors, and D_{ff} is the dimensionality of the Feed-Forward layer. The Feed-Forward Network refines these attention-derived features to form a higher-level representation of plant health, integrating information about texture, color, and shape indicative of specific stress types.

3- Layer Normalization and Residual Connections

Each sub-layer (MHSA and FFN) is followed by layer normalization and a residual connection:

$$z'_l = \mathrm{LayerNorm}\left(\mathrm{MHSA}(z_{l-1}) + z_{l-1}\right) \tag{6}$$

$$z_l = \mathrm{LayerNorm}\left(\mathrm{FFN}(z'_l) + z'_l\right) \tag{7}$$

where z_{l-1} is the output from the previous encoder layer, $MHSA(z_{l-1})$ is the result of applying multi-head self-attention to that output, z'_l is the input to the FFN sub-layer. z_l denotes the output of lth layer. These operations stabilize training and preserve fine-grained agricultural features, such as subtle color variations that may indicate early disease symptoms, ensuring that information from previous layers is not lost.

2.2.4. Classification head

The output of the Transformer encoder is a sequence of embeddings, one for each input token. For classification tasks, a special learnable classification token z_0^0 is prepended to the input sequence before encoding. After passing through all L encoder layers, the corresponding output embedding z_L^0 is used as a compact representation of the entire image. This embedding is then passed through a Multi-Layer Perceptron (MLP) to produce the final class scores:

$$\mathrm{logits} = \mathrm{MLP}\left(z_L^0\right) \tag{8}$$

where the output logits $\in \mathbb{R}^K$ (for K classes) is used for prediction. The final embedding z_L^0 aggregates the global context of all leaf patches, producing a classification output (e.g., healthy vs. diseased or specific disease types) for the entire plant image.

3. Inductive biases in ML and DL models

3.1. Definition of inductive biases

Inductive biases refer to a set of built-in assumptions that a ML model creates, which can facilitate the generalization process, especially when working with limited data. We can consider these biases as prior knowledge that the model uses to make predictions about data that it hasn't seen before. In general terms, inductive biases guide the learning algorithm to solutions that possess particular characteristics (Baxter, 2000; Goyal & Bengio, 2022).

In supervised learning, inductive biases play a crucial role in guiding models toward plausible solutions, especially when the available data is incomplete, limited, or noisy. These biases encode prior assumptions about the structure of the data or the relationships between input and output variables, enabling the model to generalize effectively beyond the training set. In CV, inductive biases often capture assumptions about spatial structures, such as the locality of visual features, spatial hierarchies, and the invariance or equivariance of patterns to transformations like translation or rotation. By embedding such priors into the model architecture or training strategy, learning becomes more data-efficient and robust to variations in real-world images (Defferrard et al., 2016; Finzi et al., 2020; Ravanbakhsh et al., 2017).

3.2. Inductive biases in CNNs and ViTs

CNNs have demonstrated remarkable effectiveness in modeling spatial hierarchies and capturing local dependencies within data. Their architecture, which consists of convolutional layers followed by pooling layers, makes them well-suited for image recognition, object detection, and segmentation, where learning spatial relationships is essential (He et al., 2015; Shelhamer et al., 2016). The efficiency of CNNs is largely attributed to the strong inductive biases natively built into their design, including locality, two-dimensional neighborhood structure, and translation equivariance (Dosovitskiy, 2020). These are uniformly applied across all layers in the network, enabling CNNs to learn invariant and spatially-conscious feature representations.

Locality and weight sharing act as key architectural constraints that reduce the number of learnable parameters and enhance generalization (d'Ascoli et al., 2021). Locality restricts each neuron's connections to a small receptive field, allowing CNNs to extract low-level features such as edges and textures, which are hierarchically combined into more abstract representations in deeper layers (Goodfellow et al., 2016). Weight sharing ensures the same filters operate across spatial locations, making the model equivariant to translation and allowing patterns recognized in one region to be detected elsewhere, thereby improving computational efficiency and generalization (Cohen & Welling, 2016). Together, these inductive biases reduce model complexity and embed prior knowledge of visual structure, improving training efficiency and performance, especially with limited labeled data. In agricultural imaging, dataset scale and diversity vary widely—from small, curated leaf datasets like PlantVillage (Mohanty et al., 2016) to large UAV or multispectral field images. CNNs, with strong biases toward locality and translation invariance, perform reliably on these agricultural datasets, particularly when data are limited or collected under controlled conditions. Owing to these characteristics, even a randomly initialized CNN can perform well in object localization tasks without training (Cao & Wu, 2022). Despite these advantages, the locality of CNNs limits their ability to capture long-range dependencies (d'Ascoli et al., 2021).

ViTs have demonstrated remarkable effectiveness in modeling spatial hierarchies and capturing global dependencies within data. Unlike CNNs, which operate locally through spatially bounded filters, ViTs rely on self-attention mechanisms where every token (or image patch) can attend to any other token regardless of spatial position (Dosovitskiy, 2020). In the ViT architecture, MLP layers function locally and preserve some translational equivariance, while self-attention layers are inherently global, integrating information across the entire spatial domain at every layer. Rather than relying on CNNs' strong inductive biases, ViTs achieve global processing through multi-head self-attention, enabling flexible, context-dependent representations. Consequently, ViTs can capture long-range relationships and broader contextual cues earlier in the network, whereas CNNs must build such spatial dependencies incrementally through repeated local convolutions, making their learning more architecture-biased than data-driven. For example, in plant disease detection, ViTs can simultaneously relate distant lesion regions across different parts of a single leaf or between multiple leaves within the same canopy, allowing the model to recognize complex and spatially dispersed disease patterns that CNNs often miss. This global contextual awareness is particularly beneficial in agricultural images where disease symptoms may appear irregularly or coexist with environmental noise such as soil, shadows, or overlapping foliage.

Interestingly, despite architectural differences, recent studies show that ViTs can implicitly learn some of the inductive biases typically found in CNNs. For example, early self-attention heads in ViTs often focus on local regions, mimicking the effect of convolutional kernels (Xiao et al., 2021). This suggests that, with sufficient training data and careful optimization, ViTs can acquire some of the useful priors that CNNs inherently possess. However, the lack of built-in inductive biases introduces challenges. Without explicit mechanisms for capturing local

details, ViTs may struggle to model fine-grained patterns, particularly in the early layers (Raghu et al., 2021). They are also highly sensitive to hyperparameter choices — such as the optimizer, learning rate schedule, and network depth — making training delicate and often requiring large datasets and substantial computational resources (Xiao et al., 2021). This trade-off between flexibility and architectural bias underscores the importance of understanding when and how to use ViTs effectively, particularly with respect to the scale and nature of the data involved.

3.3. Strategies to mitigate the lack of inductive biases in ViTs

One approach to mitigate the lack of inductive biases involves injecting convolutional inductive biases into ViTs. This can be done through convolutional patch embedding, which modifies how image data is turned into input tokens for the Transformer. Specifically, it replaces or enhances the original flat patch embedding with convolutional layers to capture local spatial patterns more effectively. Alternatively, a convolutional stem can be added at the beginning of the network to create hybrid models, helping the model learn local features more effectively.

Another strategy involves using knowledge distillation from a teacher model that possesses strong inductive biases. Knowledge distillation in neural networks entails training a smaller neural network — known as the student or distilled model — to replicate the behavior of a larger, more complex model or an ensemble of models, often referred to as the teacher or cumbersome model. The main goal is to transfer the broader understanding and generalization capability learned by the larger model into the smaller one (Hinton et al., 2015). In this approach, one or more pre-trained CNN models are often used as the teacher to transfer their inductive biases to the ViT student model (Ren et al., 2022). This method has been shown to improve the performance of ViTs, particularly on smaller datasets.

Using hierarchical structures is another way to mitigate the lack of inductive biases by introducing a multi-stage architecture that processes visual information similarly to CNNs. Instead of attending globally from the start, these models apply self-attention within local windows and gradually merge or downsample tokens across layers, creating a spatial hierarchy (Liu et al., 2021). This setup embeds locality and spatial coherence, which are key inductive biases of CNNs, allowing the model to initially capture low-level features locally and subsequently build more abstract, global representations (Wang et al., 2021). Therefore, the most common approaches used to mitigate the lack of inductive biases in ViTs can be summarized as follows:

- **Convolutional Patch Embedding:** It involves replacing simple patch embedding with convolutional layers to preserve local structures in the image, or employing recursive token aggregation, which functions similarly to convolutions in capturing local spatial relationships. This strategy is utilized in models such as Tokens-to-Token ViT (T2T-ViT) (Yuan et al., 2021).
- **Hybrid Models:** One way to mitigate the lack of inductive biases in ViTs is to use a CNN or convolutional backbone to extract low-level features and then feed this output into a ViT. This approach combines the spatial inductive biases of CNNs with the global modeling capabilities of ViTs. Introducing a few convolutional inductive biases in the early stages of ViT processing can strike a balance between inductive bias and the learning capacity of Transformer blocks, while also influencing the optimization behavior of ViTs (Xiao et al., 2021).
- **Knowledge Distillation:** Distilling knowledge from CNNs into ViTs during training is another possible solution, as exemplified by DEiT (Touvron et al., 2021), which uses a CNN as the teacher and a ViT as the student to improve performance through teacher–student learning during the training process.
- **Hierarchical Structure:** Designing ViTs with hierarchical architectures to process multi-scale features is what SwinT does, which results in improved efficiency and local feature awareness (Liu et al., 2021).

4. Applications of ViTs in precision agriculture

ViTs have been increasingly applied to plant disease detection, leveraging their powerful feature extraction and attention mechanisms. Various studies have explored different adaptations of ViTs to improve performance in this domain. These adaptations can be broadly categorized into two types: those that use ViT and its variants in their original or minimally modified forms, referred to as **Pure Models**, and those that incorporate substantial architectural modifications by integrating convolutional or other DL components, referred to as **Hybrid Models**. To provide a clear and systematic overview, this survey further organizes each category into specific sub-themes based on their methodological focus and application scope. Fig. 4 illustrates the overall taxonomy of the reviewed literature, depicting the hierarchical organization of Pure and Hybrid Models along with their corresponding subcategories.

4.1. Pure models

In this section, we review studies that employ pure ViT-based models or their minor variants designed specifically for agricultural applications. To provide a clear and organized overview, the reviewed works are grouped into four sub-themes based on their main research focus: (a) Lightweight and Efficient Designs, which aim to reduce computational complexity and model parameters for real-time or edge deployment; (b) Dataset Creation and Domain Adaptation, where researchers develop or adapt datasets to improve model robustness in agricultural environments; (c) Interpretability and Visualization, which focuses on enhancing model transparency using explainable AI tools; and (d) Transfer Learning and Few-Shot Learning Strategies, emphasizing model generalization and adaptability under limited data conditions. This thematic organization highlights not only the diversity of ViT applications in agriculture but also the evolving research priorities shaping this emerging field. In Table 2, we provide a summary of such models. Most reviewed studies employed transfer learning or fine-tuning of pretrained ViT weights (commonly initialized from ImageNet), rather than training from scratch.

(a) Lightweight and Efficient Designs: FormerLeaf (Thai et al., 2023), a Transformer-based cassava leaf disease classification model, was introduced in a recent study. The model consisted of 12 Transformer encoder layers with optimized self-attention mechanisms, which helped reduce resource usage while maintaining high classification accuracy. To enhance FormerLeaf, two optimization techniques were proposed: the Least Important Attention Pruning (LeIAP) algorithm and Sparse Matrix–Matrix Multiplication (SPMM). LeIAP identified and removed less critical attention heads, resulting in a 28% reduction in model size and a 15% increase in inference speed without significant accuracy loss. SPMM reduces computational complexity from $O(n^2)$ to $O(n^2/p)$, improving training efficiency by 10% and lowering resource consumption. These improvements made FormerLeaf suitable for real-world applications in precision agriculture, where computational efficiency is critical. The authors also mentioned challenges, including dataset imbalance and variable image quality due to diverse lighting conditions. To address these, they suggested using advanced augmentation techniques, such as GANs (Goodfellow et al., 2020), to improve model performance.

In another study, GreenViT (Parez et al., 2023b), a novel ViT-based model, was proposed to improve plant disease detection by addressing the limitations of traditional CNNs. Unlike CNNs, which may lose spatial information due to pooling layers, GreenViT segmented input images into smaller patches to effectively capture essential features. The model was evaluated using three datasets: PlantVillage (PV), Data Repository of Leaf Images (DRLI), and a new Plant Composite (PC) dataset created by merging the first two datasets. PlantVillage consisted of 54,305 images spanning 38 classes (26 infected, 12 healthy). DRLI included 4502 images of 12 plant species. GreenViT achieved

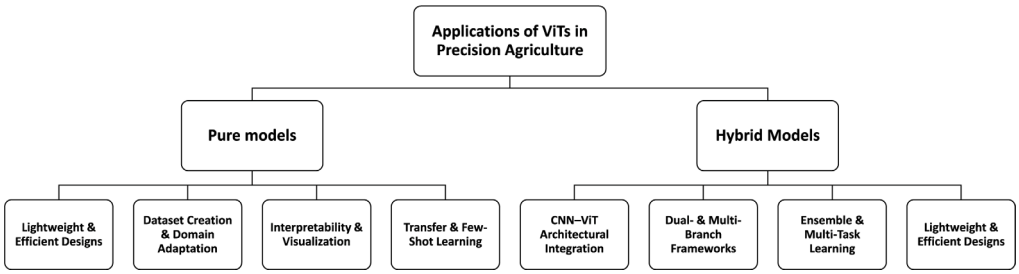

Fig. 4. Taxonomy of ViT applications in precision agriculture. The reviewed studies are organized into two main categories — Pure Models and Hybrid Models — each further divided into four sub-themes.

impressive accuracy rates of 100% on PV, 98% on DRLI, and 99% on PC, outperforming state-of-the-art models such as VGG16, MobileNetV1, and ViT-Base. The model's efficiency was highlighted by its reduced parameter count (21.65 million compared to 86 million in ViT-Base), making it computationally lightweight and suitable for edge devices. Another work introduced LeafViT (Keerthan Bhat et al., 2022), a ViT-based model tailored for detecting leaf diseases in plants, specifically targeted fine-grained features critical for classification. Experiments underscored the robustness of LeafViT across various configurations, validating its computational efficiency relative to models such as VGG16. The authors highlighted its potential for deployment on edge devices and its adaptability for object detection and segmentation tasks. Another study (Barman et al., 2024) introduced a novel application of ViT models for detecting diseases in tomato plants, leveraging smartphone-based technology. A smartphone application developed as part of that study integrated the ViT model for real-time, field-level disease detection. By deploying the model on a smartphone application via TensorFlow Lite, that work provided a practical tool for field use, enabling farmers with real-time insights to enhance crop health and yield.

In a separate work, IEM-ViT was proposed as an innovative approach to detect tea leaf diseases (Zhang, Guo, et al., 2023). It incorporated an information entropy weighting (IEW) method to prioritize regions within images that held significant information, enhancing the feature extraction process. This model used a masked autoencoder (MAE) with an asymmetric encoder–decoder architecture, effectively masking 75% of the input image to filter out redundant and irrelevant background data. This method enabled proper model development and facilitated accurate disease recognition even when low-quality images were used. The findings indicated that the IEM-ViT model surpassed previously developed algorithms. Therefore, IEM-ViT was considered suitable for fast and large-scale tea disease recognition and was proposed as deployable for recognizing diseases in other crops, contributing to enhanced agricultural management and sustainability.

(b) Dataset Creation and Domain Adaptation: The use of ViTs for plant disease detection using multispectral images captured in natural environments was explored in a recent study (De Silva & Brown, 2023). This work aimed to assess the performance of ViTs for plant disease recognition applied to multispectral imagery acquired under natural conditions. The authors created a novel dataset called the "New Multispectral Dataset", which contains images of five agricultural plants: avocados, tomatoes, limes, passionfruit, and gooseberries. The dataset includes images captured in both visible and near-infrared (NIR) modes, providing more information for disease analysis than RGB-only images. Four ViTs models, namely ViTs-S16, ViTs-B16, ViTs-L16, and ViTs-B32, were trained from scratch and subsequently fine-tuned using ImageNet-21k weights. The research examined the roles of model size, patch size, and transfer learning on the models' ability to classify diseases. The findings indicated that the ViT model based on the B16 architecture and augmented with transfer learning provided the best results, with training and testing accuracies of 93.71% and 90.02%, respectively. In another study (Bhowmik et al., 2024), authors developed a customized ViT model to identify and categorize diseases in Java Plum leaves. Their self-collected dataset consisted of six classes—Bacterial Spot, Brown Blight, Powdery Mildew, Sooty Mold, Dry, and Healthy leaves. The work demonstrated the strong potential of ViT architectures for enhancing plant disease detection, using Java Plum as an illustrative case. Notably, the proposed model outperformed traditional CNN-based approaches, achieving superior classification ac

In addition to utilizing a self-built dataset, a novel residual-distilled Transformer architecture (Zhou et al., 2023) was proposed for the identification of rice leaf diseases. The authors argued that existing works on rice leaf disease identification were unsatisfactory in terms of either accuracy or model interpretability. To address these issues, they introduced a method that combined a ViT and a distilled Transformer in a residual block for feature extraction, followed by a MLP for prediction. The researchers collected their own dataset on rice leaf disease between 2019 and 2020. This dataset, which they referred to as the rice leaf disease dataset, consisted of pictures of four rice leaf diseases: bacterial blight, brown spots, blast, and tungro. The proposed method achieved an 89% F1-score and 92% top-1 accuracy on the rice leaf disease dataset, outperforming other state-of-the-art models. An application of SwinT was introduced in a study (Zhang et al., 2021) that examined Transformer models for accurate fruit ripeness classification. The authors addressed the problem of performing fruit selection and pickup effectively in the absence of professional labor. They created their own fruit datasets, including apples and pears, to train, test, and compare various Transformer models, like ViT, SwinT, and MLP-based models. The MLP model performed well but was prone to overfitting when it had higher capacity. The moving window-based self-attention in SwinT, as shown in Fig. 5, facilitated better feature extraction of visual objects. The experiments demonstrated that SwinT yielded the best results, achieving a precision of 87.43% for fruit object detection. This work also combined the Transformer module and YOLO (Redmon et al., 2016) module to effectively classify the ripeness stages of apples and pears.

Another work (Rezaei et al., 2025) introduced a Transformer-based few-shot learning (FSL) pipeline designed to address data scarcity in plant pathology by enabling accurate disease detection from minimal annotated samples. The authors constructed a new in-field Barley dataset containing 2165 patch-level images across five classes. The images were captured under real field conditions with complex backgrounds, making the task more realistic and challenging. Their FSL framework combined transfer learning with episodic meta-training and evaluated multiple backbones, including ResNet50, ViT-B, Swin-T, and Swin-B. Using a 5-way 5-shot configuration, the model learned class prototypes from only five support samples per class and classified query images using Euclidean distance in the embedding space. The experimental results showed that Swin-B consistently provided the best feature extraction performance. On the Barley dataset, Swin-B achieved 86.78% accuracy in the 5-way 1-shot setting and 91.80% accuracy in the 5-way 5-shot setting after meta-training, outperforming the other backbones. The pipeline was also validated on a second dataset — the regenerated cassava leaf disease dataset — where Swin-B reached

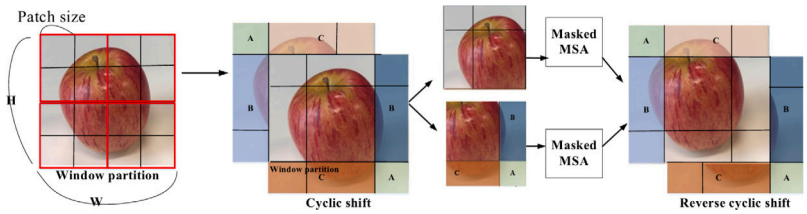

Fig. 5. The Shifted window mechanism in SwinT based on MSA (Zhang et al., 2021). The red box indicates a window where self-attention is applied, while black boxes represent individual image patches. The shaded region B illustrates the area that is masked out during shifted window self-attention to preserve the locality of each window.

97.93% (5-shot) and 95.24% (1-shot) accuracy, demonstrating strong cross-crop generalization. This study highlighted the effectiveness of Transformer-based FSL models, particularly SwinTs, for plant disease detection when only limited labeled data were available.

Another study (Ulukaya & Deari, 2025) proposed a robust ViT-based model for classifying five rice leaf diseases — blight, brown spot, caterpillar damage, fusarium, and nematode — using a newly collected in-the-wild rice leaf dataset captured under uncontrolled environmental conditions. Unlike previous works that relied on single-leaf images with uniform backgrounds, this dataset includes variations in illumination, shooting angle, distance, and complex natural backgrounds, creating a more realistic benchmark for field applications such as drone-based disease monitoring. To address the severe class imbalance, the authors applied extensive data augmentation (rotation, shearing, gamma contrast, blurring, and Gaussian noise) and experimented with categorical focal loss to improve minority-class learning. Several transfer learning baselines were evaluated, including VGG19, ResNet152, Inception V3, MobileNetV2, and EfficientNetB7, as well as ViT-B16 and ViT-B32 variants. The proposed ViT-B32 with fine-tuning, categorical focal loss, and data augmentation (ViT-B32 + FT + CFL & DA) achieved the best performance, reaching 88.57% accuracy, 91.67% specificity, and 60.00% sensitivity, outperforming all CNN-based models and other ViT variants. ViT-B32 without augmentation or CFL already surpassed the CNN baselines, confirming the advantage of Transformer-based global attention for complex, cluttered field imagery. Despite its lower accuracy compared to methods trained on clean, controlled datasets, the model demonstrates strong robustness and practical viability for drone-based or field-based early disease detection systems.

A separate study (Lin et al., 2025) proposed DIEC-ViT to address the challenges of plant disease recognition in complex real-world field environments, where background noise and significant intra- and inter-class variability often degrade model performance. Built on a ViT backbone, the model introduced two key components: the Discriminative Information Enhancement (DIE) module, which strengthened attention to foreground disease regions and suppressed irrelevant background tokens by aggregating multi-head self-attention maps from deeper layers, and the Contrastive Learning (CL) module, which restructured the feature space by reducing intra-class variability and enlarging inter-class differences. This design enabled the model to effectively capture subtle disease features while minimizing noise from natural environments. The authors evaluated the model on three real-world datasets — Soybean Disease, Paddy Disease, and PlantDoc — all containing field images with complex backgrounds, low inter-class differences, or limited samples. Experimental results showed that DIEC-ViT achieved state-of-the-art performance across all benchmarks. On the Soybean dataset, it reached 99.02% accuracy, surpassing the previous best model by 0.84%. On the Paddy dataset, the model achieved 98.94% accuracy, improving upon CNN- and Transformer-based baselines by 1%–5%, including a 3.94% improvement over a ViT-B model enhanced with an edge-feature guidance module. On the challenging PlantDoc dataset, which contained substantial intra-class variation and limited samples, DIEC-ViT obtained 78.81% accuracy, outperforming ICVT by 1.27%. Overall, DIEC-ViT effectively addressed key real-world challenges in plant disease identification and established a strong performance benchmark for ViT-based agricultural disease detection systems.

(c) Interpretability and Visualization: An improved framework of the SwinT architecture (Wang et al., 2022) was proposed to identify cucumber leaf diseases, addressing challenges such as complex backgrounds and limited sample sizes. The authors developed a SwinT-based and attention-guided GAN (STA-GAN) model capable of generating diseased images without altering background integrity. The leaf extraction module (Fig. 6), which included the proposed backbone network and Grad-CAM (Selvaraju et al., 2017), was incorporated into the GAN to create STA-GAN (Fig. 7). They employed a modified SwinT model, which used an improved feature extraction technique of step-wise small patch embeddings and Coordinate Attention (CA), without adding any extra parameters. Their results showed that STA-GAN and the improved SwinT had the potential to minimize dependency on large manually annotated datasets, while at the same time improving disease detection. In another work, LEViT (Prashanthi et al., 2024) was proposed as a model based on ViT for identifying and classifying leaf diseases in agriculture, with an emphasis on enhancing interpretability through Explainable AI techniques. Integrating Grad-CAM provided a visual explanation of the model's predictions in the form of important regions in the images of the leaves. Their model achieved high performance, with a training accuracy of 95.22%, validation accuracy of 96.19%, and test accuracy of 92.33%. Notably, Grad-CAM visualizations enhanced the trustworthiness and usability of the model by elucidating its decision-making processes. Their findings showed that LEViT's advanced structure helped to improve leaf disease recognition and diagnosis, which was essential in precision farming. The integration of interpretability enabled LEViT to address real-world issues of early disease detection and management.

Other researchers (Haque et al., 2025) introduced an enhanced ViT architecture for crop disease detection and applied it to the RicApp dataset, which contains field-conditioned images of rice and apple diseases. The proposed model incorporated a triplet multi-head attention mechanism that cascaded three attention units with residual connections inside each Transformer encoder, enabling progressive refinement of feature representations and stronger modeling of fine-grained disease patterns. The network included patch extraction, positional embedding, five Transformer encoder layers, and a classifier, totaling approximately 2.25 million parameters. Through this design, the model effectively captured both local and global relationships in images with complex backgrounds. The authors also visualized learned features using LIME and t-SNE to demonstrate interpretability and the model's ability to focus on relevant symptomatic regions. The improved model achieved 97.99% accuracy on the RicApp test set, outperforming the standard ViT by 2.2%. Evaluation on the PlantVillage dataset yielded 98.57% accuracy along with high precision and recall, confirming strong generalization to controlled-condition datasets. Comparative analysis and paired t-tests showed that improvements over competing models were statistically significant. These results highlight the robustness and efficiency of the proposed network and demonstrate its suitability for real-time disease monitoring and broader applications in precision agriculture.

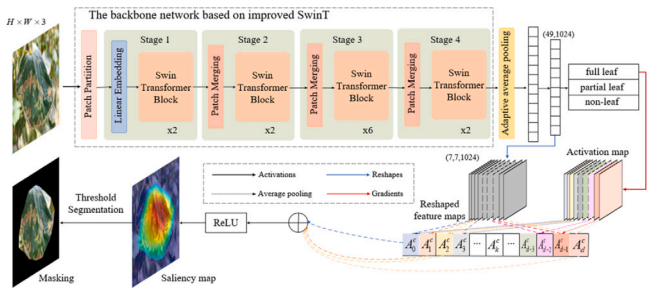

Fig. 6. Overview of the proposed leaf extraction module (Wang et al., 2022). It includes an improved SwinT-based backbone for leaf region determination, Grad-CAM for generating a saliency map, and threshold segmentation for extracting the leaf region from complex backgrounds.

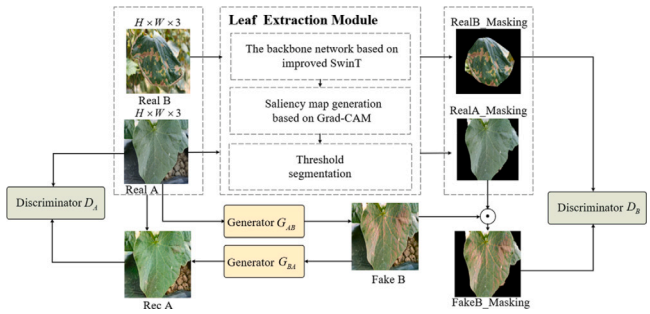

Fig. 7. The framework of the proposed STA-GAN (Wang et al., 2022). The generator transforms healthy images (A) into diseased images (B) using a series of convolutional, residual, and transposed convolutional layers. Leaf region masks guide both the generator and discriminator to focus on leaf areas, enhancing disease translation accuracy.

(d) Transfer Learning and Few-Shot Learning Strategies: An innovative transfer learning strategy designed for plant disease recognition in scenarios with limited data was proposed in a recent study (Xu et al., 2022). Pre-training was performed using the PlantCLEF2022 dataset (Joly et al., 2022), which contained 2,885,052 images across 80,000 classes. The authors employed a ViT model that was initially self-trained on ImageNet and subsequently trained on the PlantCLEF2022 dataset. Several benchmark models, such as CNNs and other ViT-based systems, were used to validate the proposed method. The results indicated that the model outperformed the previous models in all dataset configurations, including few-shot learning scenarios. For instance, it achieved remarkable results even when trained with only 20 images per class, and it also performed well in plant growth stage prediction and weed recognition. Moreover, the dual transfer learning approach improved accuracy while also demonstrating faster convergence, showcasing the strength of the method across various dataset types. This study underscored the effectiveness of large-scale, plant-related datasets combined with advanced models like ViT for transfer learning. It emphasized the necessity of appropriate source datasets, self-supervised pre-training, and fine-tuning processes for plant disease recognition. In another work (Wang et al., 2023), the SwinT network was used to improve the identification of weeds in agriculture. To address challenges posed by limited training data at scale, a two-stage transfer learning strategy was employed. First, the authors pre-trained the SwinT model on the ImageNet dataset and then fine-tuned it on the Plant Seedlings dataset as well as on a private dataset named MWFI (Maize/Weed Field Image). This strategy leveraged pre-trained network parameters to specialize in feature extraction for weed recognition. The proposed method achieved a recognition accuracy of 99.18%, outperforming other CNN-based architectures such as EfficientNetV2 (Tan & Le, 2021).

Another work (Nguyen et al., 2024) proposed a ViT-based approach for the automatic classification of strawberry diseases using transfer learning and fine-tuning techniques. The authors enhanced the ViT model by adding new layers such as ReLU activation, batch normalization, dropout, and a softmax classifier. Comparative evaluations against conventional CNN models like VGG16, VGG19 (Simonyan & Zisserman, 2014), ResNet50V2 (He et al., 2016b), and MobileNet (Howard et al., 2017) showed that the proposed ViT model, especially when combined with data augmentation, significantly outperformed other methods. In their best setting, the model achieved an accuracy and F1-score of 92.7% on the test set. The research highlighted the strengths of Transformer-based architectures in precision agriculture, especially in complex visual tasks such as disease recognition from strawberry leaves, fruits, and flowers. The study also evaluated the proposed method on the PlantVillage dataset to assess generalizability, where it attained 98.9% accuracy, outperforming both the original ViT and previous state-of-the-art methods like Mask R-CNN (He et al., 2017) and MCLCNN. The authors concluded that the ViT model's ability to capture global image features made it a powerful alternative to CNNs for agricultural disease detection tasks.

A separate study (Shehu et al., 2025) introduced three Transformer-based transfer learning models named ViT-ImageNet, ViT-Base, and ViT-Small to enhance early detection of tomato leaf diseases. The authors evaluated the models on the well-known PlantVillage dataset, which contains over 18,000 tomato leaf images across ten classes, and on a newly collected real-world dataset called TomatoEbola, which includes images from three Nigerian farms with diverse environmental conditions. Experimental results showed that these Transformer models outperformed common CNNs such as VGG19, EfficientNetB2, InceptionV3, and DMCNN. ViT-Base achieved the highest performance on the PlantVillage dataset with an accuracy of 99.17%, while ViT-Small reached 92.73% on the Kukareta subset and 91.54% on the Kasaisa subset of the TomatoEbola dataset. The findings demonstrated that Transformer-based transfer learning significantly improved robustness and reduced error rates compared to existing DL approaches, offering a more reliable solution for early and accurate tomato leaf disease diagnosis.

In summary, the development of lightweight and efficient designs in plant disease detection has shown that while these models are computationally efficient, their performance is often hindered by real-world data challenges. Factors such as dataset imbalance — where certain disease classes dominate — can bias predictions, and uneven image quality resulting from low resolution, poor lighting, or color distortion can further degrade accuracy. Moreover, interpretability and visualization techniques have become increasingly important in bridging the gap between AI-driven predictions and expert agricultural knowledge. By providing visual explanations that highlight the regions of leaves influencing model decisions, these approaches enhance transparency and build trust among agricultural practitioners. Similarly, transfer learning and few-shot learning strategies address challenges posed by limited or imbalanced datasets. Through approaches such as stratified 5-fold cross-validation, these strategies ensure that models learn balanced feature representations across all categories, improving the generalization of disease detection across various plant types and conditions. Table 3 provides a comprehensive analysis of each category, detailing their respective accuracy ranges and average accuracy values. It further identifies two common advantages and two recurring disadvantages observed across the reviewed studies.

4.2. Hybrid models

While pure ViT architectures have demonstrated strong performance in agricultural image analysis, many researchers have explored hybrid models that combine the strengths of ViTs with convolutional or other DL components. These hybrid approaches aim to balance local feature extraction with the global contextual modeling capabilities

Table 2
Summary of pure models. These models employ the pure form of ViT, aiming to enhance its performance or fine-tune it. (Reported metrics are directly extracted from the original publications. In certain studies, per-class metrics are provided; metrics that are not reported are indicated with a "–".)

Paper	Dataset(s)	Accuracy (%)	Precision	Recall	F1	Main Contribution
Thai et al. (2023)	Cassava Leaf Disease Dataset	98.52	–	–	0.9682	Proposing LeIAP algorithm for selecting key attention heads in Transformer
Parez et al. (2023b)	PlantVillage DRLI PlantComposite	100.00 98.00 99.00	– – –	– – –	– – –	Reducing number of learning parameters
Keerthan Bhat et al. (2022)	Plant Pathology 2020	97.15	Per-class	Per-class	Per-class	ViT-based approach to identify and amplify subtle discriminative regions
Barman et al. (2024)	PlantVillage	90.99	Per-class	Per-class	Per-class	Smartphone-based plant disease detection
Zhang, Guo, et al. (2023)	Tea leaves disease classification	93.78	0.9367	0.9380	0.9364	Asymmetric encoder–decoder architecture for masked autoencoder with selective patch encoding and pixel reconstruction
De Silva and Brown (2023)	Self-built	90.02	–	–	–	Using a newly collected multispectral dataset featuring five agricultural crops under real-world environmental conditions
Bhowmik et al. (2024)	Self-built	97.51	Per-class	Per-class	Per-class	Introducing customized ViT model for effective classification of Java Plum leaf diseases
Zhou et al. (2023)	Self-built	92.00	0.8800	0.9100	0.8900	Residual-distilled Transformer architecture for rice leaf disease identification
Zhang et al. (2021)	Self-built	–	0.8743	–	–	Adopt SwinT for accurate ripeness classification of various fruit types
Rezaei et al. (2025)	Self-built	97.93	–	–	–	Proposing a Transformer-based few-shot learning pipeline that enables high-accuracy barley disease detection from only a few in-field training samples using Swin-based feature extraction
Ulukaya and Deari (2025)	Self-built	88.57	–	0.6000	–	Introducing a robust ViT-based rice leaf disease classification framework that achieves state-of-the-art performance in uncontrolled field conditions
Lin et al. (2025)	Soybean Disease Paddy Disease PlantDoc	99.02 98.94 78.81	– – –	– – –	– – –	Introducing discriminative attention enhancement and contrastive feature optimization to boost ViT robustness and achieve state-of-the-art plant disease recognition in complex field environments
Wang et al. (2022)	Self-built	98.97	–	0.9873	–	Improving patch partition of SwinT and using SwinT-based Attention-guided GAN
Prashanthi et al. (2024)	New Plant Diseases Dataset	92.33	–	–	–	ViT-based model incorporating Explainable AI (XAI) features
Haque et al. (2025)	RicApp dataset PlantVillage	97.99 98.57	0.9790 0.9861	0.9794 0.9851	0.9792 0.9856	Using a triplet multi-head attention enhanced ViT to refine feature representations and achieve highly accurate crop disease detection on field images
Xu et al. (2022)	PlantCLEF2022	Varies	–	–	–	Novel transfer learning strategy for versatile plant disease recognition
Wang et al. (2023)	Plant Seedling and a self-built dataset	99.18	0.9933	0.9911	0.9922	Fine-grained recognition using SwinT and two-stage transfer learning
Nguyen et al. (2024)	PlantVillage and Strawberry Disease Detection dataset	92.70	–	–	0.9270	Fine-tuning a ViT model with additional dense and normalization layers for precise strawberry disease classification
Shehu et al. (2025)	PlantVillage Self-built	99.17 92.73	–	–	–	Developing Transformer-based transfer learning models to achieve superior accuracy and generalizability across both benchmark and real-world datasets for the early detection of tomato leaf diseases

of Transformers. To present these developments systematically, the reviewed studies are categorized into four sub-themes: (a) CNN–ViT Architectural Integration, where convolutional and Transformer layers are combined sequentially or hierarchically to enhance feature representation; (b) Dual- and Multi-Branch Frameworks, which employ parallel CNN and Transformer paths for joint learning of local and global features; (c) Ensemble and Multi-Task Learning Approaches, emphasizing cooperative model fusion or simultaneous disease and severity classification; and (d) Lightweight & Efficient Design, focusing on computationally efficient architectures for field-level or resource-constrained applications. This classification highlights how hybridization strategies extend the capabilities of ViTs, improving model robustness, scalability, and adaptability to real-world agricultural challenges. A summary of the papers is shown in Table 4. Most hybrid models were trained using transfer learning or fine-tuning strategies, leveraging pretrained ImageNet weights to overcome the limited size of agricultural datasets. In these approaches, the convolutional backbone is often frozen or partially updated during training, while the Transformer layers are optimized for agricultural image adaptation.

(a) CNN–ViT Architectural Integration: In a comparative study (Borhani et al., 2022), the authors evaluated the performance of standalone and hybrid models and proposed a approach utilizing ViT

Table 3
Summary of the characteristics of Pure model categories applied in agricultural image analysis, including their reported accuracy ranges, average accuracies, advantages, and limitations across different studies.

Subcategory	Studies	Accuracy Range (%)	Average Accuracy (%)	Advantages	Disadvantages
Lightweight & Efficient Design	Barman et al. (2024), Keerthan Bhat et al. (2022), Parez et al. (2023b), Thai et al. (2023), Zhang, Guo, et al. (2023)	90.99–100.00	96.78	•Reduced computational cost •Faster training and inference	•Sensitivity to dataset quality and imbalance •Sensitivity to hyperparameter tuning
Dataset Creation & Domain Adaptation	Bhowmik et al. (2024), De Silva and Brown (2023), Lin et al. (2025), Rezaei et al. (2025), Ulukaya and Deari (2025), Zhang et al. (2021), Zhou et al. (2023)	78.81–99.02	92.85	•Feasibility for practical agricultural automation •Enhanced Detail Recognition	•Sensitivity to Real-World conditions and environmental noise •Complexity in optimal configuration and hyperparameter tuning
Interpretability & Visualization	Haque et al. (2025), Prashanthi et al. (2024), Wang et al. (2022)	92.33–98.97	96.97	•Explainability through Grad-CAM integration •Robustness to complex backgrounds and noise	•Limited evaluation across multiple disease types and conditions •Complexity of tine-tuning and hyperparameter sensitivity
Transfer & Few-Shot Learning	Nguyen et al. (2024), Shehu et al. (2025), Wang et al. (2023), Xu et al. (2022)	92.70–99.18	95.95	•High generalization across datasets •Effective performance with limited data	•Limited interpretability and explainability •Dependence on large pretraining datasets

for automated plant disease classification. They compared ViT-based models, CNNs, and hybrid architectures that combined both techniques. These models were evaluated on three different datasets: the Wheat Rust Classification Dataset (WRCD), the Rice Leaf Disease Dataset (RLDD), and the PlantVillage dataset. A ViT-based model achieved a 100% F1-score on WRCD with a 200 × 200 resolution, outperforming EfficientNetB0 (Tan & Le, 2019) in computational efficiency. For RLDD, hybrid models balanced accuracy and computational cost, with the ViT-based model achieving the highest average precision. On the larger PlantVillage dataset, the ViT-only model reached an F1-score of 98.77%, surpassing CNN-based architectures. Models that combined attention blocks and CNNs as hybrid models offered a balance between accuracy and speed. They were comparatively faster than pure ViT models and more accurate than CNN-only models. For instance, a hybrid model achieved high accuracy in the PlantVillage experiments, indicating the potential of integrating attention mechanisms into CNNs for real-world applications. It is worth mentioning that this study highlighted the need for a practical, lightweight, real-time implementation for agricultural purposes. A separate study (Zhu et al., 2022) presented a hybrid model combining CNN with Transformer encoders to improve the accuracy and generalization of crop disease diagnosis. The authors emphasized handling complex backgrounds and disease similarities, challenges that traditional CNNs struggled to address. Using the PlantVillage dataset, the model achieved a high recognition accuracy of 99.62%. For real-world applications, datasets with complex backgrounds were also used: Dataset1 (apple, cassava, and cotton diseases) and Dataset2 (apple scab, cassava brown streak, and cotton boll blight), where the proposed model attained balanced accuracies of 96.58% and 95.97%, respectively, demonstrating robust generalization capabilities. The key to the model's success was the integration of a Transformer encoder to capture global features, which compensated for the limitation of CNN in extracting global context. Furthermore, a novel hybrid loss function, combining cross-entropy with Centerloss, was introduced to optimize feature separability and reduce intra-class variance.

ConvViT (Li et al., 2022), which combined ViT and CNNs, was developed to identify kiwifruit diseases in complex natural environments. The authors created a custom dataset of 25,168 images capturing six types of kiwifruit leaf diseases and validated the model's performance on this dataset, as well as on the publicly available PlantVillage and AIChallenger2018 datasets. To enhance feature extraction, the model employed overlapping patch embeddings and alternating convolutional and Transformer layers, ensuring both global and local feature representation. On the kiwifruit dataset, ConvViT achieved a top identification accuracy of 98.78%, surpassing benchmarks such as ResNet, ViT, and ResMLP (Touvron et al., 2022), while maintaining a lightweight design with reduced parameters and FLOPs. Key improvements included overlapping patch embedding to preserve local image continuity and reduce computational complexity, as well as a multi-stage design to balance feature extraction efficiency and computational cost. As shown in Fig. 8, the Transformer's global attention complemented CNN's local feature extraction, making ConvViT highly effective in complex environments with variable lighting, backgrounds, and noise. The model's innovative design and lightweight adaptability made it a valuable backbone for broader identification tasks in agriculture, potentially aiding real-world crop disease management efforts.

EfficientRMT-Net (Shaheed et al., 2023) is a novel hybrid model that integrates ResNet-50 and ViT to classify potato plant leaf diseases effectively. It employs depth-wise convolution for computational efficiency and utilizes a stage-block architecture to enhance scalability and feature sensitivity. This paper emphasizes that it efficiently processes high-dimensional data while maintaining high accuracy, making it a valuable tool for smart farming. Another paper (Zeng et al., 2023) proposed the SEViT model, which combines Squeeze-and-Excitation (Hu et al., 2018), ResNet101, and ViT to tackle large-scale and fine-grained plant disease classification challenges. The preprocessing network employs Squeeze-and-Excitation modules within ResNet101 to enhance channel-specific feature representation, while the ViT component leverages its global attention mechanism for feature extraction and classification. SEViT addresses limitations of CNN-based methods, which often struggle to distinguish diseases in visually similar crops. Although comparisons with popular models such as MobileNetV3 (Howard et al., 2019), EfficientNet, and VGG16 highlight SEViT's efficiency in handling visually similar diseases across different crops, it still faces challenges. These include high computational requirements due to the depth of the preprocessing network and a reliance on pretrained weights to optimize the performance of ViT.

An enhanced ViT model (Pacal, 2024), based on the MaxViT (Tu et al., 2022) architecture, was introduced for identifying diseases in maize leaves. By adapting the MaxViT structure with the addition of Squeeze-and-Excitation (SE) blocks and implementing Global Response Normalization (GRN) in the MLP layers, the proposed model achieved significant improvements in both accuracy and inference speed. The researcher created a comprehensive maize dataset by merging the

Table 4
Summary of hybrid models. These models are designed to leverage the complementary strengths of ViT and CNN architectures. (Reported metrics are directly extracted from the original publications. In certain studies, per-class or scenario-specific metrics are provided; metrics that are not reported are indicated with a "–").

Paper	Dataset(s)	Accuracy (%)	Precision	Recall	F1	Main Contribution
Borhani et al. (2022)	Wheat Rust Classification Dataset (WRCD), the Rice Leaf Disease Dataset (RLDD), and PlantVillage	Scenario-specific	Scenario-specific	Scenario-specific	Scenario-specific	Proposing lightweight DL approach based on ViT, CNN, and hybrid models for real-time automated plant disease classification
Zhu et al. (2022)	PlantVillage Custom1 Custom2	99.62 96.58 95.97	–	–	–	Proposing a hybrid model of CNN and Transformer to enhance diagnosis in complex scenes
Li et al. (2022)	PlantVillage AIChallenger2018 Self-built	99.84 86.83 98.78	0.9850 0.8534 –	0.9898 0.8342 –	0.9865 0.8539 –	Proposing a hybrid model (ConvViT) for kiwifruit disease with improved patch embedding and reduced complexity
Shaheed et al. (2023)	PlantVillage	99.24	Scenario-specific	Scenario-specific	Scenario-specific	Proposing hybrid model of ResNet-50 and ViT for classifying potato plant leaf diseases
Zeng et al. (2023)	Gathered from search engines	88.34	0.8833	0.8761	0.8750	Combining Squeeze-and-Excitation ResNet101 and ViT for large-scale and fine-grained disease classification
Pacal (2024)	Combination of PlantVillage, PlantDoc, and CD&S	99.24	0.9915	0.9937	0.9926	Adapting MaxViT structure with SE blocks and implementing Global Response Normalization in MLP layers for 4-class maize data
Guo et al. (2022)	Cucumber, Banana, Potato, Tomato datasets	Different in model variants	–	–	–	Proposing Convolutional Swin Transformer (CST) model for robust and accurate plant disease detection under natural, controlled, and noisy conditions
Salamai et al. (2023)	Paddy Doctor	98.74	0.9853	0.9786	0.9818	Introducing lesion-aware model with multi-scale contextual feature extraction and weakly supervised lesion localization
Salamai (2024)	Coffee leaf stresses dataset	99.10	–	–	0.9932	Proposing a Dual-path ViT for efficient and interpretable diagnosis
Karthik et al. (2023)	Citrus datasets	98.19	0.9839	0.9819	–	Proposing a dual-branch network combining GSDFP (Group Shuffle Depthwise Feature Pyramid) for local multi-scale feature extraction and SwinT for global context learning for citrus disease classification
Si et al. (2024)	Plant Pathology	97.32	0.9733	0.9740	0.9736	Introducing DBCoST, a dual-branch architecture that integrates CNN for extracting local features and SwinT for capturing global information, with a Feature Fusion Module (FFM) to enhance disease identification in apple leaves
Pang et al. (2023)	Custom CD&S	93.01 99.69	0.9299 0.9969	0.9301 0.9969	0.9299 0.9969	Proposing the DenseCNNs and Transformer Network (DCTN), featuring a novel multi-head self-attention mechanism for accurate detection of field crop diseases
Yang et al. (2023)	AIChallenger2018	99.00	–	–	Per-class	Proposing the TSTC network, a triple-branch Swin Transformer model for simultaneous disease and severity classification, utilizing multitask feature extraction, compact bilinear pooling, and deep supervision to enhance performance and achieve high accuracy on both tasks
Padshetty et al. (2025)	PlantVillage Wheat Leaf dataset Rice Leaf Diseases	98.95 98.89 98.80	0.9765 0.9751 0.9737	0.9635 0.9624 0.9616	0.9528 0.9511 0.9500	Using a twin-branch ViT with DS-VGG16 features, deformable attention, SwiGLU-based MLPs, and an edge-aware enhancement module to achieve highly accurate crop disease classification
Reis and Turk (2024)	Potato Leaf Dataset	99.33	Per-class	–	–	Using ensemble models for potato disease classification
Li and Tanone (2024)	PlantVillage & Kaggle	Varies by voting ways	Varies by voting ways	Varies by voting ways	Varies by voting ways	Using ensemble models for leaf disease classification

(*continued on next page*)

Table 4 (*continued*).

Paper	Dataset(s)	Accuracy (%)	Precision	Recall	F1	Main Contribution
Kalpana et al. (2024)	PlantVillage	Per-class	Per-class	Per-class	Per-class	Using ensemble of SwinT and residual CNN models
Zhang, Cen, et al. (2023)	Field-PV, PlantVillage, Syn-PV	–	Per-class	Per-class	–	Proposing Cross-Resolution Transformer (CRFormer) with large-kernel attention and multi-path Feed-Forward Network for accurate grape leaf disease segmentation in complex backgrounds
Yang, Li et al. (2024)	AIChallenger2018	Varies by plant	–	–	–	Proposing LDI-NET, a multi-label network for simultaneous identification of plant type, leaf disease, and severity using a single-branch model, combining CNN and Transformer strengths for feature extraction and employing a multi-label decoder for improved feature utilization
Tabbakh and Barpanda (2023)	PlantVillage Wheat Rust Dataset	98.81 99.86	0.9872 0.9978	0.9876 0.9965	0.9873 0.9971	Proposing hybrid model combining transfer learning and ViT for deep feature extraction and classification of plant diseases
Hu et al. (2023)	PlantVillage	99.80	–	–	0.9931	Proposing hybrid model combining adaptive Fourier Neural Operators and CNNs to enhance global and local feature extraction for plant leaf disease recognition
Shereesha et al. (2022)	Self-built	94.17	0.9334	–	0.9459	Proposing Compact Convolutional Transformer (CCT)-based model for automated detection of anthracnose and powdery mildew diseases in mango leaves
Liu et al. (2023)	Self-built	–	0.9800	0.9700	–	Proposing lightweight NanoSegmenter model based on Transformer architecture for high-precision and efficient tomato disease detection, incorporating inverted bottleneck, quantization, and sparse attention techniques
Feng et al. (2024)	PlantVillage Cassava leaves Tomato leaves	99.80 84.29 99.90	0.9998 0.8107 0.9980	0.9997 0.8205 0.9993	0.9997 0.7929 0.9990	Combining EfficientNet with group-wise Transformer for leaf disease detection
Singh et al. (2024)	PlantVillage PlantDoc	99.92 75.72	0.9960 0.7500	0.9960 0.7400	0.9950 0.7200	Proposing LeafyGAN, a GAN-based augmentation framework for synthetic leaf disease data generation, enabling lightweight MobileViT models to achieve high diagnosis accuracy even with limited real data
Li et al. (2023)	Self-built	99.10	0.9777	0.9847	0.9775	Using Multi-Head Self-Attention (MSA) and a Spatial Convolutional Self-Attention-based Transformer (SCSA-Transformer) for accurate and efficient recognition of multiple strawberry disease classes
Thakur et al. (2024)	5 Public, 1 Self-built	Varies by dataset	Varies by dataset	Varies by dataset	Varies by dataset	Proposing a real-time plant disease identification system using drone-based surveillance, featuring a lightweight ViT and CNN fusion model with conditional attention and statistical squeeze-and-excitation
Thakur et al. (2025)	5 Public	Varies by dataset	Varies by dataset	Varies by dataset	Varies by dataset	Proposing an ultra-lightweight and interpretable hybrid model that combines CNN and ViT in a parallel architecture for accurate and efficient plant disease classification, suitable for deployment on resource-constrained IoT devices
Dai et al. (2025)	5 Public , 1 Custom	Varies by dataset	Varies by dataset	Varies by dataset	Varies by dataset	Proposing a novel lightweight hybrid architecture that combines CNN and ViT components through a dual-stream token and channel mixer, along with a Bilinear Attention Transformation module for efficient local–global feature extraction

(*continued on next page*)

Table 4 (*continued*).

Paper	Dataset(s)	Accuracy (%)	Precision	Recall	F1	Main Contribution
Saranya et al. (2024)	Self-built	98.00	0.9700	0.9800	0.9700	Introducing Hybrid Pooled Multihead Attention (HPMA) model that improves pest classification by effectively capturing local and global features
Chakrabarty et al. (2024)	PlantVillage Dhan-Shomadhan	97.43 96.22	0.9700 0.9600	0.9600 0.9500	0.9600 0.9500	Introducing an interpretable hybrid model of lightweight CNN and Transformer for rice leaf disease

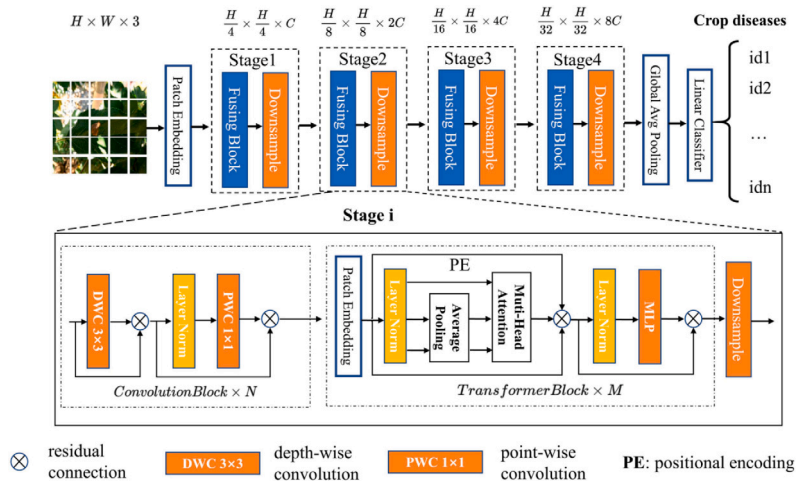

Fig. 8. Overview of the ConvViT architecture (Li et al., 2022). The top row illustrates the overall hierarchical structure of ConvViT, while the bottom row presents a detailed view of Stage 2. Each stage is composed of N convolutional blocks followed by M Transformer blocks.

PlantVillage, PlantDoc, and CD&S datasets, forming the largest publicly available maize disease dataset. The model was benchmarked against 28 CNN and 36 ViT architectures. It achieved a record-breaking accuracy of 99.24% on the test set, outperforming existing methods in both accuracy and inference speed, making it particularly well-suited for real-world agricultural applications.

In another study, a novel Convolutional SwinT (CST) model was proposed to address challenges in plant disease detection (Guo et al., 2022). By integrating convolutional layers into the SwinT architecture, the CST model enhanced feature extraction from noisy and complex agricultural images. The study evaluated the model on multiple datasets: the Cucumber Plant Diseases Dataset, Banana Leaf Disease Images, Potato Disease Leaf Dataset, and a tomato subset of PlantVillage. Notably, CST maintained an accuracy of 79.50% even with 30% salt noise, demonstrating its robustness. In addition to its strong classification performance, CST exhibited notable resilience to noise, outperforming baseline models such as ResNet50 and LeViT-192 under the same conditions. While CST surpassed existing models in accuracy on noisy data, its computational complexity — exceeding 48 million parameters — was highlighted as a limitation.

A novel approach using ViTs was introduced to address challenges in paddy leaf disease identification (Salamai et al., 2023). The model incorporated a multi-scale contextual feature extraction mechanism to capture both local and global representations of lesions on paddy leaves. This was complemented by a weakly supervised Paddy Lesion Localization (PLL) unit that prioritized significant lesion areas to enhance classification accuracy. The model's performance was evaluated on the Paddy Doctor dataset (Petchiammal et al., 2023). The proposed model outperformed several state-of-the-art models, including DenseNet169 (Huang et al., 2017), ResNet50, and SwinT, achieving an accuracy of 98.74%, an F1-score of 98.18%, and an AUC of 99.49%. These results demonstrated significant improvements over baseline models, highlighting the efficacy of lesion-focused feature extraction and context-aware learning in accurately classifying complex diseases. The authors emphasized that integrating lesion localization with multi-scale ViT structures enabled robust performance across diverse visual challenges. The model's enhanced ability to identify and prioritize discriminative lesion areas provided more reliable disease detection, supporting its potential for real-world applications in precision agriculture.

(b) Dual- and Multi-Branch Frameworks: Another way to create a hybrid model is to build a multi-stage or multi-branch model that combines CNN and ViT by dividing the network into sequential stages or branches. This hierarchical approach reduces computational costs while preserving essential information, enabling robust performance in complex environments.

Some studies utilize multi-track models, particularly dual-track (or dual-path) architectures, that integrate two parallel networks, such as a ViT and a CNN, to leverage the strengths of both. Fig. 9 shows COFFORMER (Salamai, 2024), a novel dual-path model designed for efficient and interpretable diagnosis of coffee leaf diseases, which integrates lesion segmentation and disease classification into a unified framework . The segmentation path employs a U-shaped architecture built on COFFORMER blocks, which use multiscale convolutional pooling for token mixing, while the classification path incorporates a Coffee Lesion Attention (CLA) module to focus on critical lesion regions. The framework evaluates both the type and severity of diseases, enabling real-time and explainable disease diagnosis. The dataset used for evaluation consists of 1685 images of Arabica coffee leaves captured under controlled conditions and annotated for five disease types and severity levels. Experimental results demonstrate COFFORMER's superior performance compared to state-of-the-art models. For segmentation, it achieved a Dice Similarity Coefficient (DSC) of 98.14% and a mean Intersection over Union (mIoU) of 97.98%, outperforming baselines such as Swin-Unet (Cao et al., 2022) and PSPNet (Zhao et al., 2017). In classification tasks, COFFORMER achieved an accuracy of 99.10% and an F1-score of 99.32% for disease type detection, alongside 99.12% accuracy for severity estimation. These results demonstrate its robustness in identifying diverse lesions and disease severity across real-world coffee leaf images.

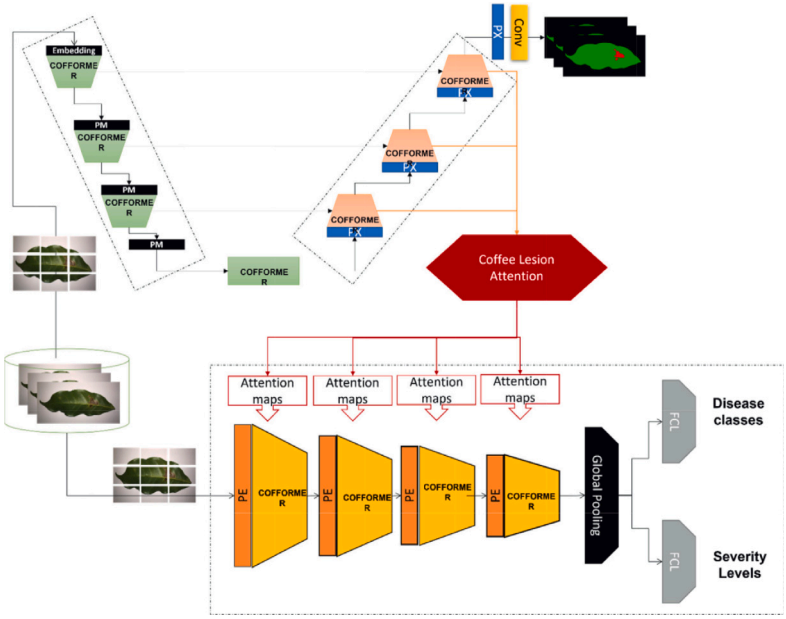

Fig. 9. The architecture of COFFORMER. It consists of three main components: a segmentation subnetwork for detecting biotic stress lesions, a dual-head classification subnetwork for identifying disease type and severity, and a CLA block that enhances feature representation using guidance from the segmentation decoder (Salamai, 2024).

For classification of citrus diseases, a novel dual-track deep fusion network (Karthik et al., 2023) was proposed that integrates a CNN-based Group Shuffle Depthwise Feature Pyramid (GSDFP) and a SwinT. In this architecture, the GSDFP branch extracts local features using convolutional layers and a Feature Pyramid Network (FPN) for multiscale feature fusion, while the SwinT branch captures global features through hierarchical feature maps and self-attention mechanisms. The outputs from these branches are fused and passed through a Shuffle Attention (SA) module to enhance contextual relationships between features. This dual-track design effectively combines local and global feature extraction, enabling precise classification. The proposed network achieved state-of-the-art results, outperforming existing models. It obtained a classification accuracy of 98.19%, surpassing benchmarks such as DenseNet-121 (92%) and AlexNet (94.3%). Another work proposed a dual-branch model, named DBCoST (Si et al., 2024), which integrates CNN and SwinT for classifying apple leaf diseases. The model includes a Feature Fusion Module (FFM) that combines information from both branches to improve accuracy. To enhance feature integration, the paper also introduces a channel attention mechanism that adjusts the importance of different feature channels. The model aims to address challenges such as noise from natural environments, including overlapping branches and fruits in the images. In comparative experiments with state-of-the-art methods, including EfficientNetV2S and MobileNetV3L, DBCoST demonstrates superior performance in recognizing apple leaf diseases, particularly in challenging natural environments.

In a recent paper, a hybrid DL model (Pang et al., 2023) was introduced to detect crop diseases in complex field environments. Recognizing the limitations of conventional CNN-based models in handling diverse real-world scenarios, the authors proposed a dual-branch network that fused CNNs for local feature extraction and Transformers for global context modeling. The architecture incorporated a four-layer pyramid structure and a novel attention mechanism, using multi-head self-attention enhanced by depthwise separable convolutions and downsampling to reduce computational overhead. This design ensured that critical disease-related areas on leaves were emphasized while mitigating the influence of noisy backgrounds. To validate their model, the authors curated a custom dataset of 45,547 images depicting twelve categories of healthy and diseased crops, captured under real-world field conditions and sourced from both competitive and open datasets. Additionally, DCTN was evaluated on the publicly available CD&S dataset, which includes maize leaf disease images. DCTN achieved state-of-the-art accuracy rates of 93.01% on their own dataset and 99.69% on the CD&S dataset, outperforming popular CNN- and Transformer-based models such as ResNet50, EfficientNetB5, and DeiT-small. These results highlighted DCTN's robustness and generalization capability in realistic agricultural settings, offering a promising direction for practical plant disease diagnostics.

The Triple-Branch SwinT Classification (TSTC) (Yang et al., 2023) network has been proposed for the simultaneous and separate classification of plant diseases and their severity. The proposed model uses a multitask feature extraction module with a triple-branch structure, integrating SwinT for feature extraction and compact bilinear pooling (CBP) for feature fusion. It also employs a deep supervision module to enhance feature discrimination across both hidden and output layers, improving the model's accuracy and generalization. Unlike single-task models, TSTC's multitask approach avoids species dependency and improves flexibility, addressing challenges like imbalanced datasets through robust feature fusion and supervision strategies. A critical contribution of this study lies in its end-to-end multitask framework, which leverages CBP to enhance feature fusion and deep supervision to improve training efficiency. The findings suggest that TSTC is highly effective in agricultural applications, providing accurate predictions that can inform targeted disease management strategies.

Another work (Padshetty et al., 2025) proposed Twin-ViT, a novel framework for accurate and efficient crop disease classification. The method combines a Depthwise Separable VGG16 (DS-VGG16) backbone for extracting both global and local features with a twin-branch Transformer architecture that incorporates deformable attention, spatial–channel attention, and a residual MLP with SwiGLU activation to enhance generalization and stability. An Edge-Aware Enhancement Module (EAEM) further emphasizes disease-relevant edge patterns, while a Squeeze-and-Excitation (SE) block in the classifier head recalibrates feature channels to improve discriminative capability. The model processes leaf images through two parallel branches that capture complementary local and global representations, which are then fused via cross-attention for robust disease identification. The framework was evaluated on three crop disease datasets: PlantVillage (using selected

classes from five crop types), the Rice Leaf Disease (RLD) dataset (containing three rice disease categories), and the Wheat Leaf (WL) dataset (featuring images of both healthy and diseased wheat). Across these datasets, Twin-ViT achieved strong performance, obtaining accuracies of 98.95% on PlantVillage, 98.89% on WL, and 98.80% on RLD, with an average accuracy of 98.88%. Comparative experiments demonstrated that Twin-ViT substantially outperformed existing methods, highlighting its effectiveness for real-world crop disease detection.

(c) Ensemble and Transfer Learning Approaches: Another idea for creating a hybrid model is using ensemble models. In a recent study (Reis & Turk, 2024), researchers introduced the MDSCIRNet architecture, a novel DL model combining Depthwise Separable Convolution (DSC) and ViT-based multi-head attention mechanisms for detecting potato leaf diseases. In addition to the standalone model, hybrid approaches integrating classical ML methods (e.g., SVM, Random Forest, Logistic Regression, and AdaBoost) with the MDSCIRNet model were proposed. These combinations yielded competitive results, particularly in cases leveraging ensemble learning techniques. In another study (Li & Tanone, 2024), the authors introduced an ensemble learning approach using hard and soft voting strategies that integrated three CNN models (MobileNetV3, DenseNet201, and ResNeXt50 (Xie et al., 2017)) and two Transformer models (ViT and SwinT) to classify leaf diseases. The experimental results showed that ensemble learning improved classification accuracy, with ViT achieving the best performance as a standalone model. The authors also employed Grad-CAM visualization to highlight the regions of the input images that were most influential in the classification decisions, demonstrating that the models effectively localized the disease areas. Their results indicated that combining CNN and Transformer models within an ensemble framework provides a powerful approach for effective plant disease classification.

A novel ensemble model, Residual Swin Transformer Networks (RST-Nets) (Kalpana et al., 2024), was proposed to address the challenges of plant disease recognition, including noise, varying image intensities, and subtle differences between healthy and diseased plants. The model integrates residual convolutional blocks with SwinT, leveraging the latter's hierarchical architecture for scalable complexity and global context awareness. The residual connections enhance the feature extraction process by retaining crucial information from earlier layers. Although the model achieved outstanding results and surpassed state-of-the-art models such as ResNet and GoogLeNet (Szegedy et al., 2015), its computational overhead limits its suitability for resource-constrained IoT and edge devices.

The CRFormer (Zhang, Cen, et al., 2023) model was introduced to address the challenge of segmenting grape leaf diseases in complex natural environments. The model employs a unique cross-resolution mechanism that maintains high-resolution (HR) and low-resolution (LR) feature maps in parallel, thereby enhancing contextual understanding. It incorporates a Large-Kernel Mining (LKM) attention mechanism for adaptive spatial and channel encoding, as well as a Multi-Path Feed-Forward Network (MPFFN) for multi-scale representation. Additionally, the model uses a lightweight Hamburger decoder to achieve effective multi-resolution data fusion. Experimental results demonstrate CRFormer's strong capability in handling the complexities of real-world agricultural segmentation tasks.

A novel multi-label model, LDI-NET (Yang, Li et al., 2024), shown in Fig. 10, was proposed for the simultaneous identification of plant type, leaf disease, and severity. The model stands out due to its single-branch architecture, which avoids complex network designs and excessive categorization. LDI-NET is built around three main modules: a feature tokenizer, a token encoder, and a multi-label decoder. The feature tokenizer integrates the strengths of both CNN and Transformers, combining CNN's ability to capture local details with the Transformer's proficiency in extracting long-range global features. This module tokenizes image data into compact spatial features, enhancing both local and global contextual awareness. The token encoder module plays a crucial role in enriching these extracted tokens by facilitating information exchange among them through multi-head self-attention and MLP structures. This design allows LDI-NET to capture complex relationships among plant type, disease, and severity features. The multi-label decoder module, which incorporates a residual structure, processes these context-rich tokens. It integrates shallow- and deep-level features through adaptive feature embeddings and cross-attention mechanisms to efficiently output multi-label identification results. The results underscore the model's ability to handle multi-label identification tasks effectively, demonstrating its potential for enhancing plant disease detection in real-world agricultural applications.

In a recent research, the authors proposed a hybrid model, TLMViT (Tabbakh & Barpanda, 2023), for plant disease classification. They incorporated data augmentation techniques, which addressed class imbalance and mitigated overfitting, along with a two-phase feature extraction process: initial extraction using pre-trained CNNs models (e.g., VGG16, VGG19, ResNet50, and AlexNet (Krizhevsky et al., 2012)), and deep feature extraction using ViT. Final classification was conducted using an MLP classifier. The model achieved a validation accuracy of 98.81% on the PlantVillage dataset and 99.86% on the Wheat Rust dataset. These results represented a 1.11% and 1.099% improvement in accuracy compared to transfer learning models without ViT. The findings indicated the effectiveness of combining transfer learning and ViT for deep feature extraction. The authors also highlighted that TLMViT effectively leveraged pre-trained models to reduce dimensionality and computational complexity. Another paper (Hu et al., 2023) proposed the FOTCA model, which combines the strengths of Transformers and CNNs for plant leaf disease detection. By integrating an Adaptive Fourier Neural Operator (AFNO) with traditional convolutional downsampling, FOTCA effectively captures both global and local features. Key innovations in FOTCA include the use of learnable Fourier features for positional encoding, which enhance feature representation by mapping images to the frequency domain. This approach improves robustness to variations such as rotation and scaling, addressing the limitations of ViT on small- and medium-sized datasets. Transfer learning using a pre-trained ImageNet model also optimizes training efficiency, enabling faster convergence.

(d) Lightweight & Efficient Design: In a recent work (Shereesha et al., 2022), an automated system was proposed for identifying two prevalent mango leaf diseases, anthracnose and powdery mildew, commonly found in the Andhra Pradesh region of India. The authors employed Compact Convolutional Transformer (CCT) models, specifically CCT-7/8 × 2 and CCT-7/4 × 2, to classify diseased leaves using DL-based ViT architectures. These models were selected for their efficiency in handling small datasets with reduced computational requirements. The experimental evaluation was conducted using a self-built mango leaf image dataset collected from over 50 trees. The dataset comprised 574 images labeled across four classes: healthy, dead, anthracnose, and powdery mildew. The study confirmed that compact Transformers like CCT are highly effective for disease detection in resource-limited agricultural contexts. In another study (Liu et al., 2023), an innovative approach for accurate tomato disease identification was presented through the development of the NanoSegmenter model, which integrates Transformer architectures with lightweight techniques such as inverse bottlenecks, quantization, and sparse attention mechanisms. This method aims to balance high precision with computational efficiency, addressing challenges in traditional models, including detailed instance segmentation and efficient deployment on mobile devices. The use of lightweight processing enables smartphone integration, which is vital for practical field use by farmers, thereby streamlining disease detection and supporting timely agricultural interventions.

The EfficientNet Convolutional Group-Wise Transformer (EGWT) (Feng et al., 2024) is a novel architecture for crop disease detection. It combines the convolutional capabilities of EfficientNet with a hierarchical and lightweight group-wise Transformer mechanism.

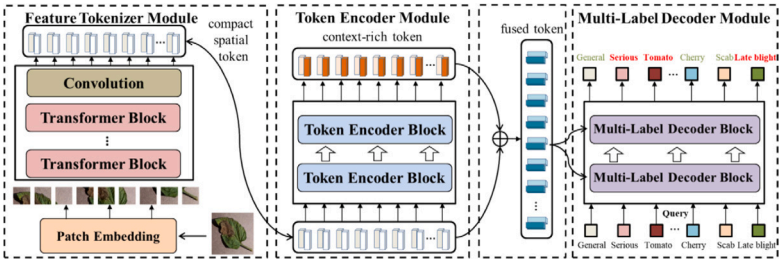

Fig. 10. The architecture of LDI-NET (Yang, Li et al., 2024). It includes a feature tokenizer module that leverages both CNN and Transformer strengths, a token encoder module for modeling relationships among plant type, disease, and severity, and a multi-label decoder for selectively extracting features to enable accurate multi-label identification.

This architecture addresses challenges related to computational efficiency and parameter constraints while maintaining high accuracy. EGWT extracts local features via EfficientNet convolutions and processes them through grouped Transformer modules, reducing redundancy and emphasizing relevant spatial relationships. The EGWT architecture was validated on three benchmark datasets: PlantVillage, Cassava, and Tomato Leaves. Visualization of intermediate layers confirmed its ability to focus accurately on disease-affected regions. Despite challenges in recognizing complex cassava leaf patterns and handling imbalanced datasets, EGWT demonstrated superior performance across several metrics. Its lightweight design makes it an excellent choice for deployment in resource-constrained environments, potentially transforming automated crop disease diagnosis in agriculture.

Another study proposed an innovative approach to enhancing plant disease diagnosis through synthetic data augmentation. The research introduced LeafyGAN (Singh et al., 2024), a GAN framework comprising a pix2pix GAN for segmentation and a CycleGAN for disease pattern generation. By generating synthetic images to address dataset imbalances, LeafyGAN ensured robust augmentation, producing high-quality representations of diseased leaves. A MobileViT classification model (Mehta & Rastegari, 2021), selected for its lightweight architecture and computational efficiency, was trained on the augmented datasets, achieving accuracies of 99.92% on the PlantVillage dataset and 75.72% on the PlantDoc dataset. These results highlight the efficacy of LeafyGAN in supporting models tailored for low-resource environments. LeafyGAN demonstrates notable improvements over existing augmentation methods, such as the LFLSeg module and traditional GANs, by preserving the background during disease pattern translation and producing visually coherent images. The integration of MobileViT, with its blend of ViT and convolutional operations, underscores a balance between performance and efficiency. Compared to resource-intensive architectures, MobileViT effectively handles the augmented datasets, demonstrating scalability for real-world applications. This framework marks a significant step toward addressing data scarcity and computational limitations, paving the way for practical deployment in agriculture for disease diagnosis and prevention.

The Spatial Convolutional Self-Attention-based Transformer (SCSA-Transformer) (Li et al., 2023) has been proposed to enhance strawberry disease identification under complex backgrounds. The research addresses challenges like class imbalance, limited large-scale datasets, and background complexity in agricultural disease detection. The SCSA-Transformer leverages convolutional layers to encode spatial features alongside a Transformer module for global feature extraction, improving efficiency and precision compared to existing models. The proposed model reduced the number of parameters by nearly half compared to the SwinT, facilitating lighter deployment and faster training. These enhancements underscore the method's applicability to real-world agricultural scenarios, particularly for identifying diseases with diverse and complex visual backgrounds.

A real-time plant disease identification framework called CondConViT (Thakur et al., 2024) has been proposed, which integrates a ViT with a Conditional Convolutional Neural Network (CondConv) and is further enhanced by a novel attention mechanism known as Conditional Attention with Statistical Squeeze-and-Excitation (CASSE). To improve generalization and robustness, the authors introduced a data augmentation technique based on a modified CycleGAN, termed C3GAN, which synthesized realistic in-field images from healthy plant samples. The model was designed to be lightweight, with only 0.95 million parameters, and was deployable on edge devices for drone-based surveillance. CondConViT was evaluated against seven state-of-the-art models and demonstrated superior interpretability and classification performance across diverse environmental conditions using both Grad-CAM and LIME techniques. They used six datasets, including five public datasets: PlantVillage, Embrapa (Barbedo et al., 2018), Plant Pathology (Thapa et al., 2020), Maize and Rice (Chen et al., 2020), and a newly created in-field drone image dataset named IIITDMJ-Maize, which comprised 416 labeled images of maize plants captured under various weather conditions. C3GAN was employed to augment the IIITDMJ-Maize dataset by generating 300 synthetic images for three maize diseases: common rust, northern leaf blight, and gray leaf spot. The CondConViT model achieved the highest or near-highest performance across all datasets in terms of accuracy, F1-score, and AUC. Furthermore, it exhibited strong generalization on unseen raw drone images and outperformed heavier models, making it suitable for real-time and resource-constrained agricultural applications.

In another study (Thakur et al., 2025), a parallel fusion of CNN and ViT modules was proposed to capture both local and global features. Unlike sequential CNN–ViT hybrids, this approach enables independent feature extraction followed by fusion using MobileNetV2 (Sandler et al., 2018) and Squeeze-and-Excitation (SE) blocks, thereby improving efficiency and interpretability. ConViTX is ultra-lightweight, comprising only 704,882 trainable parameters and requiring 0.647 GFLOPs, making it ideal for deployment on resource-constrained IoT and edge devices. The model was rigorously evaluated on five datasets: PlantVillage, Embrapa, PlantDoc, a custom in-field maize dataset (IIITDM-Maize), and a combined multi-species dataset called PlantCOMB. On PlantVillage, ConViTX achieved an accuracy of 99.63%, an F1-score of 99.64%, and an AUC of 99.96%, outperforming nine state-of-the-art models. On Embrapa, it achieved an F1-score of 95.12% and an AUC of 99.74%, while on PlantDoc, it retained the top position across most metrics. On the custom maize dataset, it achieved 93.02% accuracy, matching the best competitors while demonstrating a 2% higher AUC. On raw drone images, it achieved 61.42% accuracy, surpassing most state-of-the-art models in terms of generalizability. Beyond static evaluation, ConViTX was deployed on a drone equipped with a Jetson Nano and tested on real-time video frames captured from maize farms. It processed frames at approximately 29 FPS and demonstrated competitive performance, notably identifying northern leaf blight with 58% accuracy and healthy plants with 73% accuracy. Although challenges remain with subtle diseases such as common rust, the model achieved a better balance between speed, accuracy, and interpretability compared to other lightweight and heavyweight models.

RepAgrViT (Dai et al., 2025), a novel lightweight hybrid architecture, was proposed to integrate CNN and ViT principles for plant

disease recognition in Internet of Things (IoT) edge environments. Unlike traditional CNN–ViT hybrids, RepAgrViT employs a unique dual-stream design consisting of a token mixer and a channel mixer arranged in series to process local and global features simultaneously. A core innovation is the Bilinear Attention Transformation (BATransform) module, which captures long-range dependencies between diseased and healthy leaf regions. Additionally, the model integrates structural reparameterization techniques to reduce inference complexity while preserving accuracy. With just 9.6M parameters, the model demonstrates strong computational efficiency while retaining competitive performance. To optimize hyperparameters efficiently, the authors introduced Lite-AVPSO, a novel Particle Swarm Optimization variant that incorporates adaptive weighted delayed velocity and neighborhood-based local search mechanisms. This approach significantly enhances the hyperparameter search process compared to alternatives such as TPE, GA, and Hyperband. Through an extensive optimization process spanning over 200 trials, the model achieved a 98.54% average accuracy across three plant disease datasets: the Coffee Leaf Dataset, the Plant Pathology Dataset, and the Rice Leaf Disease Dataset. On these datasets, RepAgrViT achieved accuracies of 98.03%, 96.99%, and 98.78%, respectively.

A Hybrid Pooled Multihead Attention (HPMA) model (Saranya et al., 2024) was proposed for agricultural pest classification, enhancing both local and global feature extraction in images. HPMA integrates hybrid pooling with ViT techniques to address the limitations of conventional attention mechanisms and CNN models. It employs a novel attention mechanism that combines max and min pooling to improve feature weighting and discriminative power. The model was tested on a newly collected dataset of 10 pest classes, achieving a testing accuracy of 98%. Its effectiveness was further validated on two benchmark datasets: a medium dataset (MD) with 40 classes and a large dataset (LD) with 102 classes. For the MD, HPMA achieved a testing accuracy of 98.02%, while for the LD, it reached 95.98%. These results emphasize the model's adaptability to varying dataset sizes and complexities. Comparisons with state-of-the-art CNN and Transformer models highlight HPMA's efficiency, offering competitive accuracy with relatively lower computational requirements.

In another paper (Chakrabarty et al., 2024), researchers integrated lightweight CNNs and Transformer-based frameworks, specifically the BEiT model (Bao et al., 2021). This fusion leveraged the feature extraction strengths of CNNs and the contextual learning capabilities of Transformers to detect rice leaf diseases effectively. The model processed images by dividing them into visual tokens and applying a two-stage mechanism: reconstruction of visual elements and masked image modeling. The reconstruction stage extracted meaningful patterns, while the masked image modeling stage predicted hidden sections of the image, enhancing the model's understanding. To improve interpretability, the LIME technique was employed, coupled with segmentation using Simple Linear Iterative Clustering (SLIC), which highlighted the critical regions contributing to predictions. The explainable nature of the model, supported by attention mapping and segmentation, provided transparency in decision-making, fostering trust among end-users in agricultural diagnostics. This architecture was particularly suited for deployment on resource-constrained devices, enabling real-time application in smart agriculture.

In summary, hybrid models in agriculture demonstrate diverse architectural strategies that balance accuracy, efficiency, and interpretability across various crop and plant analysis tasks. Table 5 summarizes the characteristics of each hybrid ViT subcategory, including their accuracy ranges, advantages, and limitations, offering a comparative overview of their performance and design trade-offs within agricultural applications. Among the reviewed subcategories, CNN–ViT integrations and Dual-/Multi-Branch frameworks achieve strong average accuracies through enhanced feature extraction and attention mechanisms, proving particularly effective for handling complex agricultural imagery such as disease detection and crop classification. Ensemble and Multi-Task Learning approaches stand out for their superior accuracy and adaptability to diverse field conditions, though they require careful tuning to prevent overfitting. Meanwhile, Lightweight and Efficient hybrid ViTs emphasize real-world usability by reducing computational costs while maintaining high accuracy, making them suitable for deployment on edge devices and field-based systems. However, their dependence on data preprocessing and limited coverage of certain crop types remain challenges. Overall, these findings highlight the potential of hybrid architectures to enhance precision agriculture through improved image interpretation and decision support.

5. Findings and open challenges

5.1. Key findings

Several important findings have emerged from the review of ViTs in precision agriculture. ViTs have shown substantial improvements over traditional CNNs in various agricultural tasks, notably in plant disease detection and crop monitoring. Their ability to capture long-range relationships within images enables more accurate identification of plant diseases and pests, even in complex environments. Combining ViTs with CNNs in hybrid models has further boosted classification accuracy and robustness, making them a promising approach for agricultural challenges. Furthermore, transfer learning, in which ViTs pre-trained on large datasets are fine-tuned for agricultural applications, has proven effective in addressing data limitations and improving model performance without requiring vast amounts of labeled data. We can summarize the key findings as follows:

- **ViTs demonstrate strong performance in plant disease detection:** Many studies have shown that ViTs can achieve high accuracy in classifying plant diseases, often outperforming traditional CNN models. Their ability to learn long-range dependencies in images is particularly beneficial for plant disease detection, where symptoms can be distributed across the leaf.
- **Self-attention mechanism is a key advantage:** The self-attention mechanism allows ViTs to focus on the most important parts of an image without manual feature extraction. This is helpful in detecting small, subtle spots on leaves. ViTs can dynamically weigh different regions of the input image, ensuring that significant features receive adequate attention during classification. This helps capture both local and global contextual information. This approach is different from CNNs, which apply convolution operations uniformly across an image and can sometimes miss small but significant features. The ability of ViTs to focus on key features helps improve accuracy and reduce the need for extensive pre-processing of images.
- **ViTs can be optimized for improved performance:** Transfer learning can be effectively used to enhance the performance of ViTs for plant disease identification, even with limited data. This involves pre-training the model on a large dataset and then fine-tuning it on a smaller, disease-specific dataset. Attention head pruning can reduce model size and improve inference speed by removing less important attention heads without significant loss of accuracy. Sparse matrix multiplication in self-attention blocks can improve training efficiency and reduce GPU consumption without sacrificing performance. Knowledge distillation can be used to train a smaller, faster model that retains the performance of a larger model, which is especially useful in situations where computational resources are limited.
- **Hybrid models are highly effective in improving performance:** Combining ViTs and CNNs can leverage the strengths of both architectures. For example, CNNs, by extracting local features, can help ViTs see better, allowing the model to capture fine-grained details of the image while maintaining the ability to

Table 5
Summary of the characteristics of Hybrid model categories applied in agricultural image analysis, including their reported accuracy ranges, average accuracies, advantages, and limitations across different studies.

Subcategory	Studies	Accuracy Range (%)	Average Accuracy (%)	Advantages	Disadvantages
CNN–ViT Architectural Integration	Borhani et al. (2022), Guo et al. (2022), Li et al. (2022), Pacal (2024), Salamai et al. (2023), Shaheed et al. (2023), Zeng et al. (2023), Zhu et al. (2022)	88.34–99.62	96.76	•Improved handling of Large-Scale and Fine-Grained datasets •Effective integration of attention for discriminative feature focus	•Complexity in Model Architecture and Interpretability •Risk of overfitting and sensitivity to hyperparameters
Dual- and Multi-Branch Frameworks	Karthik et al. (2023), Padshetty et al. (2025), Pang et al. (2023), Salamai (2024), Si et al. (2024), Yang et al. (2023)	86.83–99.84	97.33	•Enhanced Feature Extraction •Robust Handling of Noisy and Complex Environments	•Limited Interpretability and Explainability •High Computational Complexity
Ensemble and Multi-Task Learning Approaches	Hu et al. (2023), Kalpana et al. (2024), Li and Tanone (2024), Reis and Turk (2024), Tabbakh and Barpanda (2023), Yang, Li et al. (2024), Zhang, Cen, et al. (2023)	98.81–99.86	99.45	•Adaptability to Real-World conditions •Support for early detection and intervention	•Potential overfitting to specific datasets •Complexity in model training and optimization
Lightweight & Efficient Designs	Chakrabarty et al. (2024), Dai et al. (2025), Feng et al. (2024), Li et al. (2023), Liu et al. (2023), Saranya et al. (2024), Shereesha et al. (2022), Singh et al. (2024), Thakur et al. (2024, 2025)	75.72–99.92	94.46	•Efficient computational complexity •Enhanced accuracy and robustness	•Dependence on data preprocessing •Restricted disease coverage

understand global relationships and context. This ultimately improves performance in tasks such as plant disease detection and image classification.

These findings highlight the potential of ViTs in precision agriculture and demonstrate their effectiveness in automated plant health monitoring systems, contributing to precision farming. These developments underscore the increasing importance of ViTs in revolutionizing precision agriculture, particularly for tasks that demand high accuracy and scalability in large-scale agricultural operations.

5.2. Open challenges

Despite the significant progress made with ViTs in precision agriculture, several challenges still hinder their wider adoption. A key issue is the absence of inductive biases in ViTs. Unlike CNNs, which take advantage of spatial hierarchies and local patterns in images through convolutional filters, ViTs do not possess these built-in inductive biases. Consequently, ViTs require large, high-quality datasets to perform well, making them data-hungry. This reliance on vast amounts of data can be a major obstacle in agriculture, where acquiring labeled datasets is both costly and time-consuming. As a result, ViTs may struggle to reach optimal performance without substantial data augmentation or pre-training on large-scale datasets like ImageNet.

Another challenge is the high computational cost and resource demands of ViTs. The self-attention mechanism that enables ViTs to capture long-range dependencies within images is computationally expensive, particularly when dealing with the high-resolution images common in agricultural applications. This leads to high memory and processing power requirements, making it difficult to deploy ViTs on resource-limited devices such as drones or edge sensors used in agricultural fields. Additionally, training ViTs can take considerably longer than training CNN-based models, which may hinder their use in real-time agricultural applications where fast responses are essential. Lastly, generalizing across varied agricultural environments presents another significant challenge. While ViTs have demonstrated strong performance on specific datasets, they may not be as effective in different climates, crop types, or farming practices. The agricultural landscape is highly diverse, and models trained on data from one region or crop type may fail to adapt to new environments with different lighting conditions, backgrounds, or plant variations. This lack of environmental robustness could limit the practical application of ViTs in precision agriculture on a global scale. Therefore, enhancing the ability of ViTs to generalize across diverse conditions is a critical area for future research. We can summarize the open challenges as:

- **Need for more comprehensive and diverse datasets:** A key limitation in current literature is the heavy reliance on the PlantVillage dataset, which primarily contains images captured under controlled conditions with uniform lighting, plain backgrounds, and isolated leaves. While useful for benchmarking, such conditions greatly simplify the disease classification task and do not reflect the visual complexity of real agricultural environments. This overreliance limits model robustness and hinders the development of ViT architectures capable of handling challenges such as occlusions, mixed lighting, soil interference, and overlapping foliage commonly encountered in open-field settings. In order to quantitatively illustrate this imbalance and its impact on reported performance, Table 6 presents the number of times each dataset has been used, along with the accuracy ranges achieved and their corresponding average accuracies. To address this gap, future research should prioritize improving dataset diversity through real-field data collection, domain adaptation, and cross-dataset evaluation. Real-world conditions involve diverse plant species, multiple disease stages, varying environmental factors, and differences in image capture (e.g., angles, distances, and resolutions), which are often missing from standard datasets. The absence of such diversity reduces model generalization to unseen scenarios. Although some studies have introduced custom datasets to mitigate this issue, collecting and annotating large-scale, high-quality field data remains expensive and time-consuming.
- **Improving the interpretability of models:** While ViTs have shown promising results, their internal workings can be less interpretable than those of traditional CNNs. The self-attention mechanisms in ViTs enable the model to learn complex relationships between different parts of an image, but it is not always clear why the model makes a particular prediction or which features are most influential in its decision-making process. This lack of interpretability can make it difficult for farmers and agronomists to trust the model or understand how to use it effectively. Some studies address this by using techniques such as Grad-CAM to visualize the areas of an image that are most important for classification. Further research is needed to develop methods that provide deeper insight into the inner workings of

Table 6
Frequency of dataset usage and corresponding performance across the reviewed studies. The table also reports the accuracy range and average accuracy achieved on each dataset. Datasets compiled by aggregating images from multiple online sources are grouped under the "Custom" category. Those collected and annotated directly by the study authors are labeled as "Self-built", while datasets without a specific name or DOI, typically obtained from Kaggle, are classified under the "From Kaggle" category.

Dataset	Description	Number of uses	Accuracy Range (%)	Average Accuracy (%)
PlantVillage (Mohanty et al., 2016)	Contains over 50,000 images of plant leaves across 14 species and 26 diseases	23	90.99–100.00	98.27
Self-built	Dataset built by the authors	15	88.57–99.18	95.58
From Kaggle	Other publicly available datasets	14	84.29–99.90	95.16
Custom	Datasets gathered from search engines, created by combining other datasets, or not easily accessible	11	88.34–99.86	95.03
PlantDoc (Singh et al., 2020)	Contains over 2000 images across 13 plant species and 17 disease classes	5	75.72–99.24	84.59
Plant Pathology2020 (Thapa et al., 2020)	Contains over 3000 high-quality real-life images of apple leaf diseases under varying conditions	4	97.15–97.32	97.26
CD&S (Ahmad et al., 2021)	Contains over 4000 corn images comprising field images and augmented images	2	99.24–99.69	99.47
Embrapa (Barbedo et al., 2018)	Contains over 56,000 images of 171 diseases and other disorders affecting 21 plant species	2	93.83–94.93	94.38
PaddyDoctor (Petchiammal et al., 2023)	Contains over 16,000 annotated paddy leaf images across 13 classes	1	98.74	98.74
Plant Seedling (Giselsson et al., 2017)	Contains over 900 unique plants belonging to 12 species	1	99.18	99.18
PlantCLEF2022 (Joly et al., 2022)	Contains over 2,000,000 images and 80,000 classes	1	73.90–99.70	95.57
DRLI (Chouhan et al., 2019)	Contains over 4000 images of healthy and diseased leaves	1	98.00	98.00
PlantComposite (Parez, Dilshad, Alanazi, & Lee, 2023)	Contains over 58,000 images of healthy and diseased leaves	1	99.00	99.00
Citrus Fruits and Leaves dataset (Rauf et al., 2019)	Contains over 600 images of citrus leaves and 150 images of citrus fruits, with five classes for each	1	98.19	98.19
Dhan-Shomadhan (Hossain, 2023)	Contains over 1100 images of five rice leaf diseases	1	96.22	96.22
Strawberry Disease dataset (Afzaal et al., 2021)	Contains 2500 images of seven types of strawberry diseases	1	92.70	92.70
Maize and Rice dataset (Chen et al., 2020)	Contains 500 rice images and 466 maize images	1	92.59–98.33	95.46

ViTs and the rationale behind their predictions, such as feature visualization and saliency maps.

- **Reducing computational costs:** Transformer-based models can be computationally expensive, requiring significant computing resources. This cost stems from the self-attention mechanism, whose complexity scales with input size, making it more difficult to train on large images and datasets. As a result, deploying these models on resource-constrained devices, such as smartphones or edge devices, can be challenging, particularly in areas with limited access to high-performance computing infrastructure. Some studies have proposed solutions such as pruning less important attention heads and using sparse matrix multiplication to reduce computational overhead and enhance practicality in real-world applications. Other approaches, including knowledge distillation and hybrid models that combine ViTs with CNNs, also aim to improve the efficiency of ViTs for deployment. Further research is needed to develop more efficient architectures and training techniques to make ViTs more accessible and suitable for practical use in plant disease detection.
- **Integrating advanced Transformer paradigms for in-field adaptation:** While ViTs have demonstrated strong performance under controlled and curated datasets, their adaptability to real-field agricultural environments remains a key challenge. Recent developments in general-purpose architectures present promising pathways toward improving model flexibility, generalization, and contextual awareness in complex outdoor conditions. For

example, CLIP-guided Continual Novel Class Discovery (CNCD) introduces a decoupled framework that leverages the generalization ability of CLIP to discover new classes without forgetting previously learned ones (Yan et al., 2025). By combining CLIP-guided semi-supervised learning, CutMix-based representation enhancement, and prototype adaptation, this approach enables models to continuously learn from unlabeled data while maintaining old knowledge. Such continual adaptation is highly relevant for agricultural applications, where new disease variants or pest types may emerge over time and require model updates without retraining from scratch. SAMT-Generator (Yang, Yang et al., 2024) employs a second-attention mechanism within a multi-stage Transformer network for image captioning tasks. Its hierarchical attention and cross-modal representation design enhance the extraction of detailed visual features and their alignment with natural language descriptions, enabling more accurate and context-rich image understanding. While originally developed for general image captioning, such an architecture can be adapted to automatically interpret and describe agricultural imagery. In precision agriculture, this approach could generate human-readable reports on crop growth, stress, or pest conditions directly from aerial or ground images. By integrating the SAMT-Generator into farm monitoring systems, producers could obtain real-time diagnostic feedback, improving decision-making in irrigation, fertilization, and disease management. DeNoising-MOT (Fu et al., 2023) addresses severe occlusions in multiple-object tracking by introducing a denoising Transformer architecture that explicitly learns to recover trajectories from noisy, occluded inputs, thereby improving object continuity and identity preservation in crowded scenes. This mechanism aligns closely with agricultural vision tasks such as fruit counting or plant tracking in dense canopies, where frequent occlusions and overlapping foliage similarly degrade the performance of conventional detectors. These models collectively highlight the potential of advanced Transformer paradigms to enhance in-field adaptability, continual learning, and robust visual understanding in precision agriculture. Future research can draw on these innovations to develop ViT frameworks capable of handling dynamic, unstructured, and evolving agricultural environments.

- **Addressing the severity of disease:** Most existing studies focus only on detecting the presence or absence of a disease, without considering its severity. However, disease severity can significantly impact crop yield, and determining it is important for developing appropriate treatment plans. More research is needed to develop methods that not only detect the presence of a disease but also assess its severity. This may involve classifying the disease into different stages or using metrics to quantify its extent. Such approaches would require modifying models to output not just a class label but also a measure of disease severity. Additionally, datasets may need to be labeled with more detailed information, including both disease classes and severity levels. Future studies should explore severity assessment methods to support more practical and informed plant disease management strategies.

These challenges highlight that, while ViTs hold great potential for plant disease detection, much work remains to ensure that they are robust, accurate, and practical for real-world applications. Addressing these challenges will be crucial for realizing the full potential of ViTs in precision agriculture.

6. Conclusion

The adoption of ViTs in agriculture represents a substantive shift from conventional DL methodologies, offering considerable potential for modeling complex visual patterns without dependence on the strong inductive biases characteristic of convolutional architectures. While CNNs have traditionally demonstrated exceptional performance in agricultural applications due to their locality and translation equivariance, ViTs present a more adaptable and scalable alternative. Their effectiveness is further enhanced when combined with specialized training strategies and hybrid architectures that strategically reintroduce inductive biases, enabling a balanced trade-off between flexibility and domain-specific robustness. Collectively, these advancements indicate that ViTs, whether as standalone models or within hybrid frameworks, are poised to play a pivotal role in the future of agricultural applications.

In this survey, we have traced the evolution of ViT-based approaches in agriculture, from vanilla Transformer models to hybrid architectures that leverage the strengths of both CNNs and Transformers. The reviewed literature demonstrates the potential of ViTs across a range of agricultural tasks, including disease identification, yield prediction, and precision agriculture, with many models outperforming or complementing traditional methods. Despite these advances, several challenges remain, including data scarcity, high computational requirements, and the need for task-specific model customization. Addressing these challenges will be crucial for fully realizing the potential of ViTs in practical agricultural applications. As the field progresses, future research should prioritize improving model efficiency, adapting models to diverse agricultural domains, and developing robust, meticulously annotated agricultural datasets.

CRediT authorship contribution statement

Saber Mehdipour: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Seyed Abolghasem Mirroshandel:** Project administration, Supervision, Writing – review & editing. **Seyed Amirhossein Tabatabaei:** Methodology, Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors utilized ChatGPT to assist in checking grammar and identifying typographical errors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

Afzaal, U., Bhattarai, B., Pandeya, Y. R., & Lee, J. (2021). An instance segmentation model for strawberry diseases based on mask R-CNN. *Sensors*, *21*(19), 6565.

Ahmad, A., Saraswat, D., & El Gamal, A. (2023). A survey on using deep learning techniques for plant disease diagnosis and recommendations for development of appropriate tools. *Smart Agricultural Technology*, *3*, Article 100083.

Ahmad, A., Saraswat, D., Gamal, A. E., & Johal, G. (2021). CD&S dataset: Handheld imagery dataset acquired under field conditions for corn disease identification and severity estimation. arXiv preprint arXiv:2110.12084.

Bao, H., Dong, L., Piao, S., & Wei, F. (2021). Beit: Bert pre-training of image transformers. arXiv preprint arXiv:2106.08254.

Barbedo, J. G. A., Koenigkan, L. V., Halfeld-Vieira, B. A., Costa, R. V., Nechet, K. L., Godoy, C. V., Junior, M. L., Patricio, F. R. A., Talamini, V., Chitarra, L. G., et al. (2018). Annotated plant pathology databases for image-based detection and recognition of diseases. *IEEE Latin America Transactions*, *16*(6), 1749–1757.

Barman, U., Sarma, P., Rahman, M., Deka, V., Lahkar, S., Sharma, V., & Saikia, M. J. (2024). Vit-SmartAgri: vision transformer and smartphone-based plant disease detection for smart agriculture. *Agronomy*, *14*(2), 327.

Baxter, J. (2000). A model of inductive bias learning. *Journal of Artificial Intelligence Research*, *12*, 149–198.

Bhowmik, A. C., Ahad, M. T., Emon, Y. R., Ahmed, F., Song, B., & Li, Y. (2024). A customised vision transformer for accurate detection and classification of java plum leaf disease. *Smart Agricultural Technology*, *8*, Article 100500.

Borhani, Y., Khoramdel, J., & Najafi, E. (2022). A deep learning based approach for automated plant disease classification using vision transformer. *Scientific Reports*, *12*(1), 11554.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, *33*, 1877–1901.

Buja, I., Sabella, E., Monteduro, A. G., Chiriacò, M. S., De Bellis, L., Luvisi, A., & Maruccio, G. (2021). Advances in plant disease detection and monitoring: From traditional assays to in-field diagnostics. *Sensors*, *21*(6), 2129.

Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, Q., & Wang, M. (2022). Swin-unet: Unet-like pure transformer for medical image segmentation. In *European conference on computer vision* (pp. 205–218). Springer.

Cao, Y. H., & Wu, J. (2022). A random cnn sees objects: One inductive bias of cnn and its applications. In *Proceedings of the AAAI conference on artificial intelligence*: *Vol. 36*, (pp. 194–202).

Chakrabarty, A., Ahmed, S. T., Islam, M. F. U., Aziz, S. M., & Maidin, S. S. (2024). An interpretable fusion model integrating lightweight CNN and transformer architectures for rice leaf disease identification. *Ecological Informatics*, *82*, Article 102718.

Chen, J., Chen, J., Zhang, D., Sun, Y., & Nanehkaran, Y. A. (2020). Using deep transfer learning for image-based plant disease identification. *Computers and Electronics in Agriculture*, *173*, Article 105393.

Chouhan, S. S., Singh, U. P., Kaul, A., & Jain, S. (2019). A data repository of leaf images: Practice towards plant conservation with plant pathology. In *2019 4th international conference on information systems and computer networks* (pp. 700–707). IEEE.

Cohen, T., & Welling, M. (2016). Group equivariant convolutional networks. In *International conference on machine learning* (pp. 2990–2999). PMLR.

Dai, G., Tian, Z., Wang, C., Tang, Q., Chen, H., & Zhang, Y. (2025). Lightweight vision transformer with lite-AVPSO hyperparameter optimization for agricultural disease recognition. *IEEE Internet of Things Journal*.

d'Ascoli, S., Touvron, H., Leavitt, M. L., Morcos, A. S., Biroli, G., & Sagun, L. (2021). Convit: Improving vision transformers with soft convolutional inductive biases. In *International conference on machine learning* (pp. 2286–2296). PMLR.

De Silva, M., & Brown, D. (2023). Plant disease detection using vision transformers on multispectral natural environment images. In *2023 international conference on artificial intelligence, big data, computing and data communication systems* (pp. 1–6). IEEE.

Defferrard, M., Bresson, X., & Vandergheynst, P. (2016). Convolutional neural networks on graphs with fast localized spectral filtering. In *Advances in neural information processing systems*: *Vol. 29*.

Dosovitskiy, A. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929.

Feng, J., Ong, W. E., Teh, W. C., & Zhang, R. (2024). Enhanced crop disease detection with EfficientNet convolutional group-wise transformer. *IEEE Access*, *12*, 44147–44162.

Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. *Computers and Electronics in Agriculture*, *145*, 311–318.

Finzi, M., Stanton, S., Izmailov, P., & Wilson, A. G. (2020). Generalizing convolutional neural networks for equivariance to lie groups on arbitrary continuous data. In *International conference on machine learning* (pp. 3165–3176). PMLR.

Fu, T., Wang, X., Yu, H., Niu, K., Li, B., & Xue, X. (2023). Denoising-mot: Towards multiple object tracking with severe occlusions. In *Proceedings of the 31st ACM international conference on multimedia* (pp. 2734–2743).

Garg, T., Dwivedi, P., Mishra, M. K., Joshi, N. C., Shrivastava, N., & Mishra, V. (2024). Artificial intelligence in plant disease identification: Empowering agriculture. In *Methods in microbiology*: *Vol. 55*, (pp. 179–193). Elsevier.

Giselsson, T. M., Jørgensen, R. N., Jensen, P. K., Dyrmann, M., & Midtiby, H. S. (2017). A public image database for benchmark of plant seedling classification algorithms. arXiv preprint arXiv:1711.05458.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT press Cambridge.

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2020). Generative adversarial networks. *Communications of the ACM*, *63*(11), 139–144.

Goyal, A., & Bengio, Y. (2022). Inductive biases for deep learning of higher-level cognition. *Proceedings of the Royal Society A*, *478*(2266), Article 20210068.

Guo, Y., Lan, Y., & Chen, X. (2022). CST: Convolutional Swin Transformer for detecting the degree and types of plant diseases. *Computers and Electronics in Agriculture*, *202*, Article 107407.

Haque, M. A., Deb, C. K., Gole, P., Karmakar, S., Dheeraj, A., Shah, M. U. D., Dutta, S., Kumar, M. P., & Marwaha, S. (2025). An enhanced vision transformer network for efficient and accurate crop disease detection. *Expert Systems with Applications*, Article 127743.

He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 2961–2969).

He, K., Zhang, X., Ren, S., & Sun, J. (2015). Spatial pyramid pooling in deep convolutional networks for visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *37*(9), 1904–1916.

He, K., Zhang, X., Ren, S., & Sun, J. (2016a). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).

He, K., Zhang, X., Ren, S., & Sun, J. (2016b). Identity mappings in deep residual networks. In *European conference on computer vision* (pp. 630–645). Springer.

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531.

Hossain, M. F. (2023). Dhan-Shomadhan: A dataset of rice leaf disease classification for Bangladeshi local rice. arXiv preprint arXiv:2309.07515.

Howard, A., Sandler, M., Chu, G., Chen, L. C., Chen, B., Tan, M., Wang, W., Zhu, Y., Pang, R., Vasudevan, V., et al. (2019). Searching for mobilenetv3. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 1314–1324).

Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. arXiv preprint arXiv:1704.04861.

Hu, B., Jiang, W., Zeng, J., Cheng, C., & He, L. (2023). FOTCA: hybrid transformer-CNN architecture using AFNO for accurate plant leaf disease image recognition. *Frontiers in Plant Science*, *14*, Article 1231903.

Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7132–7141).

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4700–4708).

Jackulin, C., & Murugavalli, S. (2022). A comprehensive review on detection of plant disease using machine learning and deep learning approaches. *Measurement: Sensors*, *24*, Article 100441.

Joly, A., Goëau, H., Kahl, S., Picek, L., Lorieul, T., Cole, E., Deneu, B., Servajean, M., Durso, A., Glotin, H., et al. (2022). Overview of lifeclef 2022: an evaluation of machine-learning based species identification and species distribution prediction. In *International conference of the cross-language evaluation forum for European languages* (pp. 257–285). Springer.

Kalpana, P., Anandan, R., Hussien, A. G., Migdady, H., & Abualigah, L. (2024). Plant disease recognition using residual convolutional enlightened Swin transformer networks. *Scientific Reports*, *14*(1), 8660.

Karthik, R., Hussain, S., George, T. T., & Mishra, R. (2023). A dual track deep fusion network for citrus disease classification using group shuffle depthwise feature pyramid and Swin Transformer. *Ecological Informatics*, *78*, Article 102302.

Keerthan Bhat, H., Mukund, A., Nagaraj, S., & Prakash, R. (2022). LeafViT: Vision transformers-based leaf disease detection. In *International conference on innovations in computational intelligence and computer vision* (pp. 85–102). Springer.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*: *Vol. 25*.

Li, X., Chen, X., Yang, J., & Li, S. (2022). Transformer helps identify kiwifruit diseases in complex natural environments. *Computers and Electronics in Agriculture*, *200*, Article 107258.

Li, G., Jiao, L., Chen, P., Liu, K., Wang, R., Dong, S., & Kang, C. (2023). Spatial convolutional self-attention-based transformer module for strawberry disease identification under complex background. *Computers and Electronics in Agriculture*, *212*, Article 108121.

Li, L. H., & Tanone, R. (2024). Ensemble learning based on CNN and transformer models for leaf diseases classification. In *2024 18th international conference on ubiquitous information management and communication* (pp. 1–6). IEEE.

Li, L., Zhang, S., & Wang, B. (2021). Plant disease detection and classification by deep learning—a review. *IEEE Access*, *9*, 56683–56698.

Lin, J., Chen, X., Lou, L., You, L., Cernava, T., Huang, D., Qin, Y., & Zhang, X. (2025). DIEC-ViT: Discriminative information enhanced contrastive vision transformer for the identification of plant diseases in complex environments. *Expert Systems with Applications*, *281*, Article 127730.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 10012–10022).

Liu, Y., Song, Y., Ye, R., Zhu, S., Huang, Y., Chen, T., Zhou, J., Li, J., Li, M., & Lv, C. (2023). High-precision tomato disease detection using NanoSegmenter based on transformer and lightweighting. *Plants*, *12*(13), 2559.

Mahlein, A. K. (2016). Plant disease detection by imaging sensors–parallels and specific demands for precision agriculture and plant phenotyping. *Plant Disease*, *100*(2), 241–251.

Mehta, S., & Rastegari, M. (2021). Mobilevit: light-weight, general-purpose, and mobile-friendly vision transformer. arXiv preprint arXiv:2110.02178.

Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, *7*, 1419.

Nguyen, H. T., Tran, T. D., Nguyen, T. T., Pham, N. M., Nguyen Ly, P. H., & Luong, H. H. (2024). Strawberry disease identification with vision transformer-based models. *Multimedia Tools and Applications*, *83*(29), 73101–73126.

Pacal, I. (2024). Enhancing crop productivity and sustainability through disease identification in maize leaves: Exploiting a large dataset with an advanced vision transformer model. *Expert Systems with Applications*, *238*, Article 122099.

Pacal, I., Kunduracioglu, I., Alma, M. H., Deveci, M., Kadry, S., Nedoma, J., Slany, V., & Martinek, R. (2024). A systematic review of deep learning techniques for plant diseases. *Artificial Intelligence Review*, *57*(11), 304.

Padshetty, S., et al. (2025). A novel twin vision transformer framework for crop disease classification with deformable attention. *Biomedical Signal Processing and Control*, *105*, Article 107551.

Pang, D., Wang, H., Ma, J., & Liang, D. (2023). DCTN: a dense parallel network combining CNN and transformer for identifying plant disease in field. *Soft Computing*, *27*(21), 15549–15561.

Parez, S., Dilshad, N., Alanazi, T. M., & Lee, J. W. (2023). Towards sustainable agricultural systems: A lightweight deep learning model for plant disease detection. *Computer Systems Science and Engineering*, *47*(1), 515–536.

Parez, S., Dilshad, N., Alghamdi, N. S., Alanazi, T. M., & Lee, J. W. (2023). Visual intelligence in precision agriculture: Exploring plant disease detection via efficient vision transformers. *Sensors*, *23*(15), 6949.

Petchiammal, Kiruba, B., Murugan, & Arjunan, P. (2023). Paddy doctor: A visual image dataset for automated paddy disease classification and benchmarking. In *Proceedings of the 6th joint international conference on data science & management of data (10th ACM IKDD cODS and 28th COMAD)* (pp. 203–207).

Prashanthi, B., Krishna, A. P., & Rao, C. M. (2024). LEViT-leaf disease identification and classification using an enhanced Vision transformers (ViT) model. *Multimedia Tools and Applications*, 1–32.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al. (2019). Language models are unsupervised multitask learners. *OpenAI Blog*, *1*(8), 9.

Raghu, M., Unterthiner, T., Kornblith, S., Zhang, C., & Dosovitskiy, A. (2021). Do vision transformers see like convolutional neural networks? *Advances in Neural Information Processing Systems*, *34*, 12116–12128.

Rauf, H. T., Saleem, B. A., Lali, M. I. U., Khan, M. A., Sharif, M., & Bukhari, S. A. C. (2019). A citrus fruits and leaves dataset for detection and classification of citrus diseases through machine learning. *Data in Brief*, *26*, Article 104340.

Ravanbakhsh, S., Schneider, J., & Poczos, B. (2017). Equivariance through parameter-sharing. In *International conference on machine learning* (pp. 2892–2901). PMLR.

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 779–788).

Reis, H. C., & Turk, V. (2024). Potato leaf disease detection with a novel deep learning model based on depthwise separable convolution and transformer networks. *Engineering Applications of Artificial Intelligence*, *133*, Article 108307.

Ren, S., Gao, Z., Hua, T., Xue, Z., Tian, Y., He, S., & Zhao, H. (2022). Co-advise: Cross inductive bias distillation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16773–16782).

Rezaei, M., Diepeveen, D., Laga, H., Gupta, S., Jones, M. G., & Sohel, F. (2025). A transformer-based few-shot learning pipeline for barley disease detection from field-collected imagery. *Computers and Electronics in Agriculture*, *229*, Article 109751.

Salamai, A. A. (2024). Towards automated, efficient, and interpretable diagnosis coffee leaf disease: A dual-path visual transformer network. *Expert Systems with Applications*, Article 124490.

Salamai, A. A., Ajabnoor, N., Khalid, W. E., Ali, M. M., & Murayr, A. A. (2023). Lesion-aware visual transformer network for Paddy diseases detection in precision agriculture. *European Journal of Agronomy*, *148*, Article 126884.

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4510–4520).

Saranya, T., Deisy, C., & Sridevi, S. (2024). Efficient agricultural pest classification using vision transformer with hybrid pooled multihead attention. *Computers in Biology and Medicine*, *177*, Article 108584.

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (pp. 618–626).

Shaheed, K., Qureshi, I., Abbas, F., Jabbar, S., Abbas, Q., Ahmad, H., & Sajid, M. Z. (2023). EfficientRMT-Net—An efficient ResNet-50 and vision transformers approach for classifying potato plant leaf diseases. *Sensors*, *23*(23), 9516.

Shehu, H. A., Ackley, A., Marvellous, M., & Eteng, O. E. (2025). Early detection of tomato leaf diseases using transformers and transfer learning. *European Journal of Agronomy*, *168*, Article 127625.

Shelhamer, E., Long, J., & Darrell, T. (2016). Fully convolutional networks for semantic segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *39*(4), 640–651.

Shereesha, M., Hemavathy, C., Teja, H., Reddy, G. M., Kumar, B. V., & Sunitha, G. (2022). Precision mango farming: Using compact convolutional transformer for disease detection. In *International conference on innovations in bio-inspired computing and applications* (pp. 458–465). Springer.

Shoaib, M., Shah, B., Ei-Sappagh, S., Ali, A., Ullah, A., Alenezi, F., Gechev, T., Hussain, T., & Ali, F. (2023). An advanced deep learning models-based plant disease detection: A review of recent research. *Frontiers in Plant Science*, *14*, Article 1158933.

Si, H., Li, M., Li, W., Zhang, G., Wang, M., Li, F., & Li, Y. (2024). A dual-branch model integrating CNN and Swin Transformer for efficient apple leaf disease classification. *Agriculture*, *14*(1), 142.

Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.

Singh, D., Jain, N., Jain, P., Kayal, P., Kumawat, S., & Batra, N. (2020). PlantDoc: A dataset for visual plant disease detection. In *Proceedings of the 7th ACM IKDD coDS and 25th COMAD* (pp. 249–253).

Singh, A. K., Rao, A., Chattopadhyay, P., Maurya, R., & Singh, L. (2024). Effective plant disease diagnosis using Vision Transformer trained with leafy-generative adversarial network-generated images. *Expert Systems with Applications*, Article 124387.

Sladojevic, S., Arsenovic, M., Anderla, A., Culibrk, D., & Stefanovic, D. (2016). Deep neural networks based recognition of plant diseases by leaf image classification. *Computational Intelligence and Neuroscience*, *2016*(1), Article 3289801.

Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going deeper with convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1–9).

Tabbakh, A., & Barpanda, S. S. (2023). A deep features extraction model based on the transfer learning model and vision transformer "tlmvit" for plant disease classification. *IEEE Access*, *11*, 45377–45392.

Tan, M., & Le, Q. (2019). Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning* (pp. 6105–6114). PMLR.

Tan, M., & Le, Q. (2021). Efficientnetv2: Smaller models and faster training. In *International conference on machine learning* (pp. 10096–10106). PMLR.

Thai, H. T., Le, K. H., & Nguyen, N. L. T. (2023). FormerLeaf: An efficient vision transformer for Cassava LeafDisease detection. *Computers and Electronics in Agriculture*, *204*, Article 107518.

Thakur, P. S., Chaturvedi, S., Khanna, P., Sheorey, T., & Ojha, A. (2024). Real-time plant disease identification: Fusion of vision transformer and conditional convolutional network with C3GAN-based data augmentation. *IEEE Transactions on AgriFood Electronics*.

Thakur, P. S., Chaturvedi, S., Seal, A., Khanna, P., Sheorey, T., & Ojha, A. (2025). An ultra lightweight interpretable convolution-vision transformer fusion model for plant disease identification: ConViTX. *IEEE Transactions on Computational Biology and Bioinformatics*.

Thapa, R., Zhang, K., Snavely, N., Belongie, S., & Khan, A. (2020). The plant pathology challenge 2020 data set to classify foliar disease of apples. *Applications in Plant Sciences*, *8*(9), Article e11390.

Touvron, H., Bojanowski, P., Caron, M., Cord, M., El-Nouby, A., Grave, E., Izacard, G., Joulin, A., Synnaeve, G., Verbeek, J., et al. (2022). Resmlp: Feedforward networks for image classification with data-efficient training. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(4), 5314–5321.

Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., & Jégou, H. (2021). Training data-efficient image transformers & distillation through attention. In *International conference on machine learning* (pp. 10347–10357). PMLR.

Tu, Z., Talebi, H., Zhang, H., Yang, F., Milanfar, P., Bovik, A., & Li, Y. (2022). Maxvit: Multi-axis vision transformer. In *European conference on computer vision* (pp. 459–479). Springer.

Ulukaya, S., & Deari, S. (2025). A robust vision transformer-based approach for classification of labeled rices in the wild. *Computers and Electronics in Agriculture*, *231*, Article 109950.

Upadhyay, A., Chandel, N. S., Singh, K. P., Chakraborty, S. K., Nandede, B. M., Kumar, M., Subeesh, A., Upendar, K., Salem, A., & Elbeltagi, A. (2025). Deep learning and computer vision in plant disease detection: a comprehensive review of techniques, models, and trends in precision agriculture. *Artificial Intelligence Review*, *58*(3), 1–64.

Vaswani, A. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*.

Wang, F., Rao, Y., Luo, Q., Jin, X., Jiang, Z., Zhang, W., & Li, S. (2022). Practical cucumber leaf disease recognition using improved Swin Transformer and small sample size. *Computers and Electronics in Agriculture*, *199*, Article 107163.

Wang, W., Xie, E., Li, X., Fan, D. P., Song, K., Liang, D., Lu, T., Luo, P., & Shao, L. (2021). Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 568–578).

Wang, Y., Zhang, S., Dai, B., Yang, S., & Song, H. (2023). Fine-grained weed recognition using Swin Transformer and two-stage transfer learning. *Frontiers in Plant Science*, *14*, Article 1134932.

Xiao, T., Singh, M., Mintun, E., Darrell, T., Dollár, P., & Girshick, R. (2021). Early convolutions help transformers see better. In *Advances in neural information processing systems*: *Vol. 34*, (pp. 30392–30400).

Xie, S., Girshick, R., Dollár, P., Tu, Z., & He, K. (2017). Aggregated residual transformations for deep neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1492–1500).

Xu, M., Yoon, S., Jeong, Y., & Park, D. S. (2022). Transfer learning for versatile plant disease recognition with limited data. *Frontiers in Plant Science*, *13*, Article 1010981.

Yan, Q., Yang, Y., Dai, Y., Zhang, X., Wiltos, K., Woźniak, M., Dong, W., & Zhang, Y. (2025). CLIP-guided continual novel class discovery. *Knowledge-Based Systems*, *310*, Article 112920.

Yang, B., Li, M., Li, F., Wang, Y., Liang, Q., Zhao, R., Li, C., & Wang, J. (2024). A novel plant type, leaf disease and severity identification framework using CNN and transformer with multi-label method. *Scientific Reports*, *14*(1), 11664.

Yang, B., Wang, Z., Guo, J., Guo, L., Liang, Q., Zeng, Q., Zhao, R., Wang, J., & Li, C. (2023). Identifying plant disease and severity from leaves: A deep multitask learning framework using triple-branch Swin transformer and deep supervision. *Computers and Electronics in Agriculture*, *209*, Article 107809.

Yang, X., Yang, Y., Ma, S., Li, Z., Dong, W., & Woźniak, M. (2024). SAMT-generator: A second-attention for image captioning based on multi-stage transformer network. *Neurocomputing*, *593*, Article 127823.

Yuan, L., Chen, Y., Wang, T., Yu, W., Shi, Y., Jiang, Z. H., Tay, F. E., Feng, J., & Yan, S. (2021). Tokens-to-token vit: Training vision transformers from scratch on imagenet. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 558–567).

Zeng, Q., Niu, L., Wang, S., & Ni, W. (2023). SEViT: a large-scale and fine-grained plant disease classification model based on transformer and attention convolution. *Multimedia Systems*, *29*(3), 1001–1010.

Zhang, X., Cen, C., Li, F., Liu, M., & Mu, W. (2023). CRFormer: cross-resolution transformer for segmentation of grape leaf diseases with context mining. *Expert Systems with Applications*, *229*, Article 120324.

Zhang, Z., Gong, Z., Hong, Q., & Jiang, L. (2021). Swin-transformer based classification for rice diseases recognition. In *2021 international conference on computer information science and artificial intelligence* (pp. 153–156). IEEE.

Zhang, J., Guo, H., Guo, J., & Zhang, J. (2023). An information entropy masked vision transformer (iem-vit) model for recognition of tea diseases. *Agronomy*, *13*(4), 1156.

Zhao, H., Shi, J., Qi, X., Wang, X., & Jia, J. (2017). Pyramid scene parsing network. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2881–2890).

Zhou, C., Zhong, Y., Zhou, S., Song, J., & Xiang, W. (2023). Rice leaf disease identification by residual-distilled transformer. *Engineering Applications of Artificial Intelligence*, *121*, Article 106020.

Zhu, W., Sun, J., Wang, S., Shen, J., Yang, K., & Zhou, X. (2022). Identifying field crop diseases using transformer-embedded convolutional neural network. *Agriculture*, *12*(8), 1083.